

Locating and Steering Refusal Beyond Attention

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Abstract

Where inside a language model does refusal live, and does that place change when the architecture does? In a transformer, refusal is governed by a single direction in the residual stream, a finding that safety and interpretability tooling now depend on. State-space models (SSMs) route information through a recurrent update instead of attention. They share no token-mixing mechanism with a transformer. Does the same safety representation survive this shift, or must it be rediscovered for each new architecture? It survives. We find that a single rotation aligns one model’s representation space with another’s. Under this rigid rotation, which can only reorient a space and not reshape it, the alignment shows the two models genuinely share the representation. A harm probe trained on a transformer then flags an SSM’s harmful inputs, and removing the aligned direction makes a model answer attacks it would otherwise refuse, while a random direction of the same size does far less. What is architecture-specific is not where the direction is steered but where it must be read. Each layer computes a fresh output that is then added into the residual stream, and harm is cleanly readable at this output, the write site, before the addition. A control that holds the intervention’s strength fixed shows that what matters is where the direction is estimated, not where it is applied: a direction estimated at the write site controls refusals, while one estimated in the summed stream does not. Applied through a detector-triggered gate (a lightweight classifier flagging harmful prompts), this direction lowers jailbreak success in all four architecture families we test (SSM, transformer, recurrent, and hybrid). On the SSM it holds against an attacker that tunes its prompt against the defense: the classifier fires on all 100 attacked prompts and attack success stays at 1.0% (from 15.0%). The defense only matches, and does not beat, a trivial rule that returns a fixed refusal whenever the same detector fires, so the gate adds no defense strength of its own. The contribution is that the direction it amplifies transfers across architectures. Where target labels are scarce, a transported direction used as a harm detector (on our Llama-to-Falcon-Mamba pair) flags 11.4 percentage points more of the harmful prompts than one trained from the same few labels, at a matched 5.0% false-alarm rate. Two human raters validate the automated judge behind these numbers, agreeing with it 94.8% of the time ($\kappa=0.90$), and the judge’s few mistakes only overstate attacks, so our defense numbers are not inflated. Safety tooling built on refusal therefore ports to a new architecture by re-estimating the direction at that architecture’s write site, not by rebuilding it.

1 Introduction

The safety of a deployed language model rests on its refusals: it is safe only if it refuses what it should not answer, the one behavior every jailbreak targets. In a transformer, refusal is

governed by a single linear direction in the residual stream, and a growing safety stack reads and steers that direction to detect and block harm (Arditi et al. 2024; Zou et al. 2023a; Lee et al. 2025). Every one of these results is in a transformer, where information between positions is routed by attention. State-space models (SSMs) such as Mamba (Gu and Dao 2023; Zuo et al. 2024) route it through a recurrent state update instead, and are now entering deployment. If refusal is a property of attention, this stack breaks whenever a model routes information differently, and each new architecture needs its safety tooling rediscovered from scratch. Is refusal a property of the transformer *architecture*, or the same *function* every model learns, and where inside a model does it live?

We find that it is the function. The refusal representation is shared across architectures that route context in completely different ways: by attention (Llama-3.1, Mistral), by a selective scan (Falcon-Mamba), by token-shift recurrence (RWKV-6), or by a mix of attention and scan (Zamba2). A refusal direction read in a transformer and mapped into a Mamba state space is still the state-space model’s refusal trigger, and removing it there jailbreaks the model (Table 4). We also find that what is architecture-specific is not *where* refusal is steered but *where it must be read*: harm is cleanly decodable at the fresh contribution each architecture writes before the residual accumulates it (the attention output in a transformer, the selective-scan block output in an SSM; Figures 1 and 3 sketch this per architecture), and a matched-magnitude control shows the direction estimated there controls refusal while one estimated in the accumulated residual degenerates. This yields one rule: *refusal is architecture-independent as a representation and architecture-specific as a readout site*. We study this in Falcon-Mamba-7B, with Llama-3.1-8B (Grattafiori et al. 2024) and Mistral-7B (Jiang et al. 2023) as transformer references, and confirm it on RWKV-6 (Peng et al. 2024) and the hybrid Zamba2-7B (Glorioso et al. 2024). We make three contributions.

- **Refusal is a shared representation across architectures.** A single rigid (Procrustes) rotation (Schönemann 1966) both aligns the representation spaces and causally transports the refusal direction. Removing the transported direction jailbreaks the target: decisively across the transformer-SSM pair and at the hybrid Zamba2-7B’s Mamba write site, more weakly at its attention site and on the single-checkpoint RWKV-6. A refusal-orthogonal random direction has a far smaller effect. This is the paper’s central result.
- **The refusal direction must be read at the fresh write,**

not the accumulated residual. Harm peaks in decodability at each architecture’s write site: a linear probe separates harmful from benign prompts with AUROC 0.99 (1.0 is perfect separation, 0.5 chance) on the surface-controlled subset, where a surface-only probe scores 0.54. A matched-magnitude control (Table 7) shows the direction *estimated* there controls refusal while one estimated in the residual degenerates, and that the intervention’s *location* is irrelevant.

- **Applied through a detector-triggered gate, that direction causally controls refusal in all four architecture classes.** The gate is a causal test of the readout claim, not a defense contribution: its numbers only match a simpler baseline that returns a fixed refusal whenever the same detector fires, so what ports across architectures is the direction, not defense strength (Section 5). The gate lowers prefix-injection attack success on Falcon-Mamba (98.0% to 5.0%) and on three further classes (the two transformers, the recurrent RWKV-6, and the hybrid, Table 6). On Falcon-Mamba it also holds under a suffix optimized through the gate: the detector fires on all 100 attacked prompts and attack success stays at 1.0% (from 15.0%), where the same adaptive attack erodes a transformer’s detector.

2 Method

2.1 Threat model

An adversary controls the input prompt and may adapt to the defense, including optimizing the prompt against the frozen detector (Zou et al. 2023b; Andriushchenko, Croce, and Flammarion 2025). The adversary cannot change the model weights or the detector. The defender runs the detector and the gate at inference time. We separate two evaluations (Section 3): transfer to unseen attack families, and robustness to an attacker that adapts to the detector.

2.2 Detector and the detect-then-gate loop

The detector is a linear probe read at the routing layer, the write-site layer whose activations carry the per-prompt decision (selected in Section 2.5), one dot product per prompt, trained on pairs of harmful and benign prompts. Both the harmful and the benign prompts also appear wrapped in jail-break templates during training, so the probe cannot learn to fire on the wrapper text itself and must read the underlying request (Appendix A.29). It scores the prompt once: if $p_{\text{harm}} > \theta$ the gate is applied at the architecture’s write site for the whole generation, otherwise the model runs unmodified. The decision is made per prompt, not per token. Tokens generated after a flagged prompt also score as harmful, so a per-token gate would keep re-triggering and drift all generations toward refusal. The write-site result holds under a nonlinear (multi-layer perceptron, MLP) detector as well as a linear one, and every cell of the baseline comparison shares the same fixed detector and threshold (Appendix A.31).

2.3 Write site: the SSM selective-scan gate (anchor)

Falcon-Mamba-7B, our most fully instrumented model, is the *anchor* we dissect first. In a selective state-space model, information propagation is controlled by three input-dependent selection parameters: a step size (Δ), an input-selection gate (B), and a readout (C). Two additional matrices are fixed and learned once during training: a state-transition matrix (A) and a skip-connection matrix (D). At each layer ℓ the mixer maps the input x_t to the selection (Δ_t, B_t, C_t) and updates the hidden state via the scan $h_t = \exp(\Delta_t A) h_{t-1} + (\Delta_t B_t) x_t$, with readout $y_t = C_t h_t + D x_t$. Our decodability study (probing each layer with a linear classifier and measuring how well it separates harmful from benign activations) identifies the step size Δ , the input-dependent selection with no counterpart in a transformer, as a location of harm. The probe detects harm at Δ nearly as well as at the mixer output (Section 5), but Δ is where harm is read, not where the gate steers: both the detector and the gate act at the block output. This is the per-layer contribution y_t that the Δ -gated scan writes before it is added to the residual stream, a distinct site from the residual stream itself. We call this site the block output (equivalently the mixer output, the mixer’s out-projection in Figure 1). On a positive prompt-level decision the gate amplifies the refusal direction $\hat{u}_\ell$ at this selective-scan site for the whole generation:

$$g(y_t) = y_t + \alpha \hat{u}_\ell, \quad (1)$$

which extends the additive steering of Arditì et al. (2024) to the SSM’s selective-scan block output (formal definitions in Appendix A.1). The direction $\hat{u}_\ell$ is the mean difference between harmful and harmless activations at layer ℓ , not the detector’s discriminative direction, which reads harm well but does not control refusal when amplified (Appendix A.30). The strength α and the routing layers are set by a small held-out sweep, with α kept below the value at which generation degenerates (the coherent gain, Appendix A.25). *To our knowledge, no prior steering or safety method acts at the selective-scan block of an SSM* (Section 6).

2.4 Write site: the transformer attention output (reference)

For the transformer reference we estimate the refusal direction at the attention output projection (the fresh write) and, as a baseline, in the residual stream, and apply the same gate $g(\cdot)$ from Equation 1 at the layer where the direction induces refusal, triggered by the same prompt-level detector. The direction estimated at the attention output defends, while the one estimated in the accumulated residual degenerates, for both transformers we test, Llama-3.1-8B and Mistral-7B (Table 6; full sweep in Appendix A.24). A matched-absolute-magnitude control isolates the cause. With the same direction held fixed, pushing it at the residual and pushing it at the attention output lower attack success by statistically indistinguishable amounts (overlapping 95% confidence intervals). What matters is therefore where the direction is *estimated*, not where the gate is *applied* (Appendix A.35). The position-wise MLP output, which does not mix information

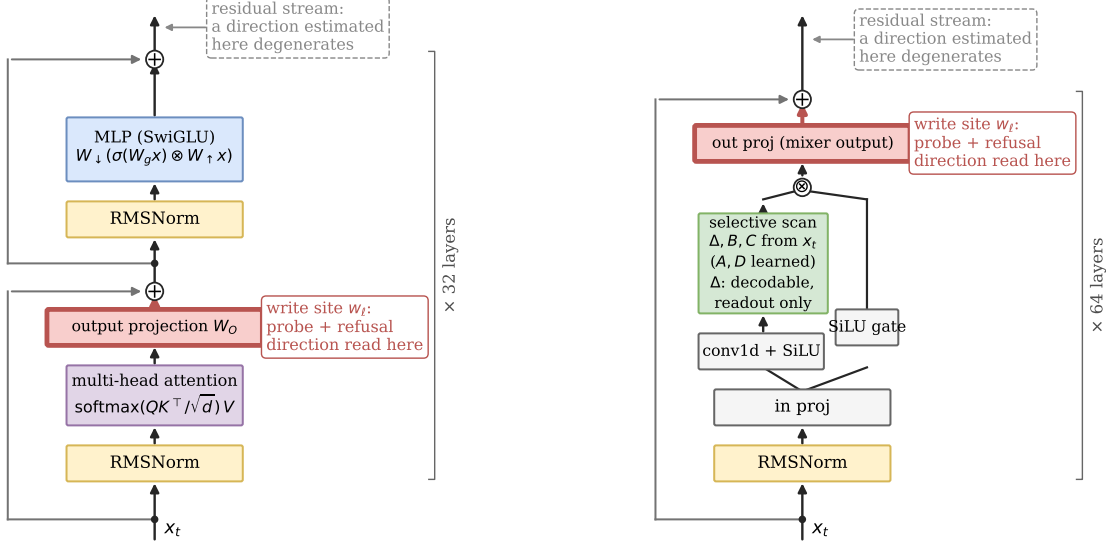


Figure 1: The write site in the two reference architectures. Left: transformer block (Llama-3.1, Mistral). The write site is the attention output projection W_O , the fresh contribution attention writes into the residual stream. Right: Falcon-Mamba block (selective-scan SSM). The write site is the mixer’s out-projection, the block output the Δ -gated scan writes before the residual accumulates it, and Δ inside the scan is decodable but a readout only. In both, the probe and the refusal direction are estimated at the red site; a direction estimated in the accumulated residual (gray) degenerates.

across positions, yields an equally usable direction, ruling out cross-position mixing as the source (Appendix A.18, A.19).

2.5 A recipe for siting the gate on a new model

These findings give a procedure for deploying the gate on a new architecture, given only a small contrastive set: find the write site where the refusal direction is cleanly decodable (the selective-scan block output in an SSM, the attention output in a transformer), select the gate layers by held-out decodability, estimate the mean refusal direction and a linear trigger there, calibrate the gain below the degeneration cliff, and deploy closed-loop. The only model-specific work is layer selection and gain calibration, where a naive transfer of one model’s constants to another fails (Appendix A.2). Which component exposes a decodable direction changes with the architecture (a pure SSM has no MLP, and in Falcon-Mamba the newly written, non-accumulated part of the block output is the skip term Dx_t , Appendix A.19), so the recipe’s first step, finding the write site, is done per architecture rather than assumed.

2.6 Baselines

The baselines vary the intervention against our gate: residual-subspace projection (LEACE-style rank- k erasure (Belrose et al. 2023)) is an open-loop residual edit, conditional activation steering (CAST) (Lee et al. 2025) shares our detector but reads and steers at the residual, and an always-on version of our push (as DeTAM does (Li, Jiang, and Wei 2025)) drops the loop (results in Table 9). All cells share the critical layers and detector threshold θ from our decodability study, with

the erasure rank k and gate strength α swept and calibrated to stay coherent (Appendix A.25).

2.7 Why projection fails and gating should not

Projection is brittle because a fixed subspace can be routed around downstream. A detection-triggered gate instead acts on the routing decision for the whole generation, leaving no fixed subspace to route around. We measure the in-distribution, out-of-distribution, and adaptive cases separately.

3 Experimental Setup

Table 1 summarizes models (including Qwen2.5-7B (Qwen Team 2025)), corpora, attacks, judges, metrics, and baselines at a glance.

The setup serves two questions, whether refusal is the same representation across the architectures and where that representation is read and steered. The detector is trained on harm versus benign-but-spicy prompts, a different split from the refusal direction’s harm-versus-Alpaca (Taori et al. 2023) fit. Decodability is reported as AUROC, the probability that a randomly chosen harmful prompt scores above a randomly chosen benign one (1.0 is perfect separation, 0.5 is chance), and uncertainty on attack-success and over-refusal rates as Wilson 95% confidence intervals, the standard interval for binomial proportions. Code, data, and result files will be released publicly upon publication.

Table 1: Experimental setup at a glance. The minimal-pair corpus is matched to XSTest in template and length. Over-refusal is measured on XSTest’s benign-but-spicy prompts (safe requests worded to sound harmful). Attack success is the Llama-Guard-3-8B-judged unsafe fraction ($n=200$ harmful, 100 benign-but-spicy per condition, 95% CI, four-judge agreement 72.8%). Capability is accuracy with and without the gate.

Models	Corpora	Attacks	Judges & metrics	Baselines
Falcon-Mamba-7B (SSM anchor) Falcon3-Mamba-7B (2nd SSM) Llama-3.1-8B, Mistral-7B, Qwen2.5-7B (transformers) RWKV-6-World-7B (recurrent) Zamba2-7B (hybrid)	AdvBench (harm) Alpaca (benign) XSTest (surface control, over-refusal)	Prefix injection (fluent) Persona roleplay (DAN, fluent) Fictional framing (fluent) Dilution (adaptive) Optimized adversarial suffix (GCG; on the SSM both a gradient-free variant and GCG through the differentiable scan)	Llama-Guard-3-8B (primary) +Llama-3.1-8B-Instruct, Mistral-7B-Instruct, ShieldGemma-9B MMLU, TruthfulQA (capability) Probe AUROC (decodability)	Residual-subspace projection CAST Perplexity filter SmoothLLM Cross-architecture linear-map transfer

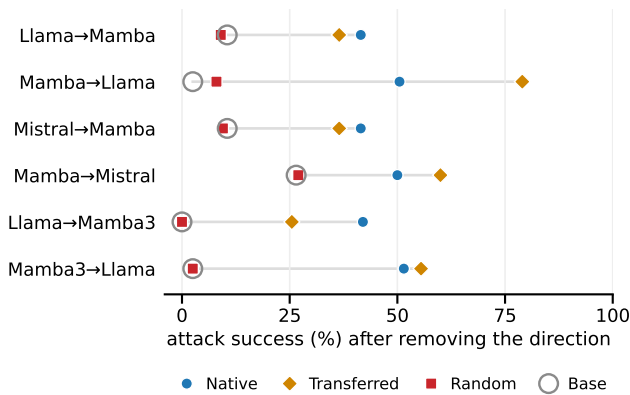


Figure 2: The refusal direction is shared across architectures. Reading it in one architecture, mapping it into another, and removing it there (Transferred) raises attack success as much as removing the target’s own direction (Native), while an orthogonalized random direction through the same map (Random) does not, in all six transfer directions. A plot of Table 5 (Mamba=Falcon-Mamba, Mamba3=Falcon3-Mamba).

4 Locating refusal across architectures

In a transformer, refusal is a direction in the residual stream. Does a pure SSM, which shares no token-mixing mechanism with attention, carry the *same* refusal representation up to a change of basis, or learn its own? We locate where the refusal signal lives in the SSM and find it is the same representation: a single rigid rotation carries it from the transformer’s activation space into the SSM’s, where it still causally controls refusal.

4.1 Harm is decodable at the selective-scan step size

Harm is linearly decodable from the Falcon-Mamba selective-scan state, concentrated in the step size Δ and the mixer output. The harm probe is tested on a version of XSTest (Röttger et al. 2024) where harmful and benign prompts are

Table 2: Where harm lives in the selective scan. Surface-controlled held-out AUROC of a linear harm probe at each selective-scan component (Falcon-Mamba-7B), on the surface-matched XSTest contrast where a surface probe is near chance (0.54). Harm is decodable at the step size Δ and the mixer output, not at B or C .

Selective-scan component	AUROC (surface-controlled)
Δ (step size)	0.95
B (input selection)	0.79
C (readout selection)	0.78
mixer output	0.99

matched for surface features like sentence structure and wording (the matched pairs are generated from a hand-written template bank crossed with length-matched harmful/benign action pairs, shipped as code in the release). A surface-only probe scores near chance there (AUROC 0.54), while the harm probe still separates them, so it is reading content rather than surface cues (per-component AUROC in Table 2). On that subset Δ is the most decodable *selection* component (0.95, against 0.79 and 0.78 for the input and readout selections B and C), and the mixer output the scan writes is more decodable still (0.99). Δ is the input-dependent gate that sets how much each token writes into the recurrent state, so harm is exposed in the SSM’s *selection* and then written at the mixer output, where decodability peaks. The defense in Section 5 reads and steers there. This also settles Δ ’s causal status: it is decodable, but a readout rather than the controlling site.

The locus is stable across model scale, with harm decodable at Δ from AUROC 0.89 to 0.99 across 5 base Mamba models from 130M to 7B parameters (Appendix A.4), and the signal is present in the transformer reference, where harm is decodable from the Llama-3.1-8B residual stream at AUROC 1.00. The finding also holds on a newer SSM generation: Mamba-2 (Codestral-Mamba-7B) shows the same decodability at its mixer output (AUROC 1.00), so it is not specific to Mamba-1 (Appendix A.10).

4.2 The refusal representation is shared, up to a rigid rotation

Transferring a steering direction within one architecture through a learned map is established (Oozeer et al. 2025). But a flexible map can synthesize the target’s own refusal axis, so it cannot by itself show the representations are shared. We test two criteria while transferring from attention to the selective scan: a structural one, whether a rigid rotation aligns the two representations, and a causal one, whether a transferred direction still carries its refusal role.

Setup. On a shared set of harmful and harmless prompts we collect paired activations from the source and target models and fit a ridge-regularized linear map W between their hidden spaces. We map the source refusal direction (mean harmful minus harmless) through W and steer the target with it, testing every architecture pair in both directions. Two behavioral tests follow, amplifying the mapped direction on jailbroken prompts (defense) and removing it from harmful prompts (ablation, where higher attack success means the direction was the refusal trigger). As a control, we repeat both with a random source direction mapped through W and orthogonalized to the target’s refusal axis, so the map cannot leak refusal signal into it (Appendix A.6).

Shared up to rotation, not merely mappable. Because a flexible map could synthesize the target’s own refusal axis, we test with an *orthogonal* (Procrustes) map, which can only rotate and has no freedom to synthesize a direction. The transported probe reaches held-out AUROC 1.00 on the unrestricted harmful/benign corpus (AdvBench versus Alpaca, where surface cues help) and 1.00 on the surface-matched minimal pairs, matching the target’s own probe (1.00), both near ceiling. A harder test confirms this isn’t just a ceiling effect. When the hardest XSTest category is held out entirely, both the target’s own probe and the transported probe drop together (0.78 vs. 0.75), tracking each other closely (Table 3). The transported probe beats a shuffled-category control in all 16 of 16 folds tested (Appendix A.8, Figure 4). A flexible map that had merely found the target’s own axis would not track this. Fitting the rotation on deliberately mismatched prompt pairs, source and target activations from different prompts rather than the same one, collapses the transported probe to AUROC 0.33 (Appendix A.7), at and below chance. Only the correct pairing transports, so the representation is genuinely shared.

This extends to a third, architecturally distinct model. RWKV-6 (Peng et al. 2024) uses token-shift recurrence, not a selective scan, yet a Llama-read harm probe transports into it at AUROC 1.00, above the shuffled control 0.43. The transport is also causal, though more weakly, with a margin narrower than the pure SSM’s or the hybrid’s Mamba site’s (Table 4; single-seed numbers, Wilson CIs, and judging methodology in Appendix A.16). The site relocates. When each site is pushed at the same fraction of its own activation norm, the residual stays inert (47.5% unchanged) while the time-mix write point steers attack success to 17.5% against 82.5% for a matched random direction, isolating direction from scale (the deployed defense is the Table 6 row).

Table 3: Per-pair category-held-out cross-validation results. Mean own and mean transported probe scores are averaged across the 8 held-out categories. Pearson r measures how closely they track each other. Full setup and interpretation in Section 4, plotted in Figure 4.

Source → target	Mean own	Mean transp.	Pearson r [95% CI]	Folds > null (shuffled-category control)
Llama-3.1-8B → Falcon-Mamba	0.95	0.95	0.99 [0.95, 1.00]	8/8
Mistral-7B → Falcon-Mamba	0.95	0.95	0.99 [0.97, 1.00]	8/8

Table 4: Removing the rotation-transported refusal direction raises attack success in every non-transformer architecture we test, while a matched random direction through the same map does much less (Llama-Guard-judged; RWKV-6 and Zamba2 pooled over three seeds). The margin is decisive on the two Mamba write sites, narrower on RWKV-6, whose random control alone jailbreaks a third of prompts, and not decisive at Zamba2’s attention site. Per-seed numbers, 95% confidence intervals (CIs), and coherence screens in Appendix A.9, A.16, and A.22.

Target (site)	Source	Base	Removed	Random
Falcon-Mamba (block out)	Llama	10.5%	44.0%	9.0%
Falcon-Mamba (block out)	Mistral	10.5%	45.0%	9.5%
RWKV-6 (time-mix out)	Llama	3.0%	54.7%	32.3%
Zamba2 (Mamba out)	Llama	8.9%	93.3%	22.8%
Zamba2 (attn. out)	Llama	8.9%	16.7%	8.9%

Transport beats retraining when target labels are scarce. Portability has practical value only if a direction carried over from the source model outperforms one fitted directly in the target. We give both the same budget of k labeled *target* prompts, spent either on the paired activations that fit the map or on the target’s own mean-difference direction, and score AUROC on a held-out harm category neither has seen. Budgets are matched in target labels only: the transport arm also carries a direction estimated on the source model’s abundant labels, which is the asymmetry the practical case turns on. Transport wins at every budget from $k=4$ to 64, by +0.032 to +0.046 AUROC, with the clustered interval excluding zero at every budget (Appendix A.41). A deployed gate runs at a threshold, where the margin is wider. Allowing 5.0% false alarms on benign prompts, the transported direction catches 74.2% of harmful prompts against 62.8% for one fit from the same labels, a gain of 11.4 percentage points. We do not convert this into a label-equivalence factor, because the from-scratch curve flattens and is not monotone in k (Appendix A.41). The gain disappears in distribution, where both arms saturate. What portability buys is therefore generalization: the transported direction catches kinds of harm that the target’s own few labels do not cover.

The shared direction carries its causal role across all six transfer directions. Removing the mapped refusal direction from harmful prompts raises the target’s attack success far above an orthogonalized random control (Table 5), across three model pairs and both directions. The rigid rotation, not just the flexible map, produces the same causal effect: the orthogonal map that structurally aligned the two models also

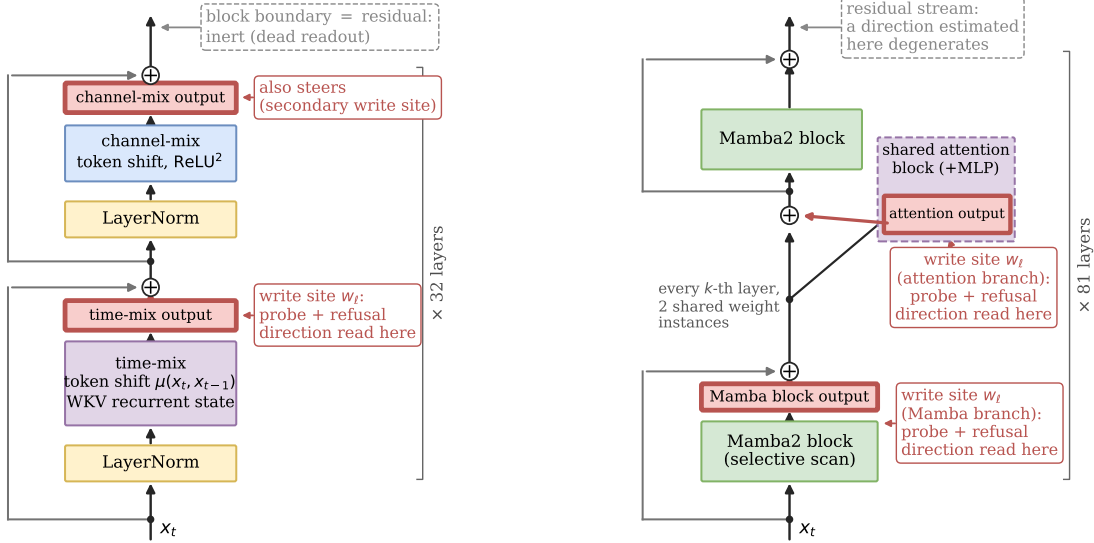


Figure 3: The write site in the two further architecture classes. Left: RWKV-6 block (token-shift RNN). The write sites are the sub-block outputs, the time-mix output (primary) and the channel-mix output (secondary), while the block boundary is architecturally the residual and is inert. Right: Zamba2 (hybrid). A Mamba2 backbone taps shared attention blocks (two weight instances reused across depth) every k -th layer, and both branch outputs are write sites.

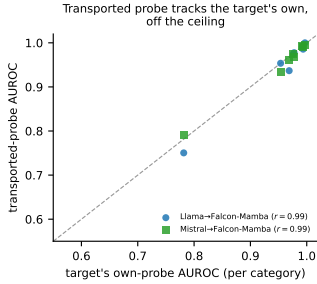


Figure 4: Category-held-out cross-validation. Each point is one held-out XSTest category, plotting the transported orthogonal probe’s AUROC against the target’s own-probe AUROC. Points follow the $y=x$ line as the own probe falls off ceiling, so the transport tracks the target’s harm structure category by category rather than sitting at a saturated ceiling.

transports the refusal direction into Falcon-Mamba. Removing that transported direction jailbreaks the SSM as much as removing its own native direction (44.0% against 41.5%), and the output stays coherent. The same rigid-rotation ablation runs in every non-transformer target we have (Table 4). One rigid rotation therefore both proves the representations are shared and controls behavior once transported (Appendix A.9).

The site where the representation transports is also where the direction is cleanly read. A source model’s refusal

Table 5: Cross-architecture transfer, ablation test (plotted in Figure 2). Removing the mapped direction (Transferred) raises attack success like removing the target’s own (Native). The Random control does not. Random is mapped through W like Transferred, then orthogonalized to the target’s refusal axis (cosine ≈ 0), so it shares the map without inheriting the refusal direction. The effect holds in all six rows: three model pairs (Mamba=Falcon-Mamba, Mamba3=Falcon3-Mamba), each tested in both directions.

Pair / direction	Base	Native	Transferred	Random
Llama→Mamba	10.5%	41.5%	36.5%	9.0%
Mamba→Llama	2.5%	50.5%	79.0%	8.0%
Mistral→Mamba	10.5%	41.5%	36.5%	9.5%
Mamba→Mistral	26.5%	50.0%	60.0%	27.0%
Llama→Mamba3	0.0%	42.0%	25.5%	0.0%
Mamba3→Llama	2.5%	51.5%	55.5%	2.5%

direction transports into the write site as the same representation, and the direction read there controls behavior when amplified, while Δ is decodable but does not control refusal (Section 5). Amplifying a model’s refusal direction read at its write site lowers prefix-injection attack success across four architecture classes (Table 6), while a direction read from the accumulated residual degenerates. A matched-magnitude control (Appendix A.35) shows the gap is the estimated direction, not the push location, matching prior reports of residual-steering brittleness (Li et al. 2026; Kornikov et al. 2025; Xiong et al. 2026), with the effect size

tracking each model’s baseline alignment. The defense also holds against a harder, gradient-optimized attack. A per-prompt greedy coordinate gradient (GCG) suffix (Zou et al. 2023b), an adversarial string optimized by gradient search to force compliance, reaches 41.0% [31.9, 50.8] attack success on Mistral-7B. The attention write site more than halves it, to 18.0% [11.7, 26.7]. ShieldGemma-9B, an out-of-family judge reading the model’s whole response rather than only checking whether it opens with GCG’s affirmative target, confirms the same pattern (16.0% to 5.0%, Appendix A.24). The hybrid Zamba2-7B corroborates the pattern. A Llama-trained harm probe transports into it cleanly under the same rigid rotation (AUROC 1.00 vs a shuffled control 0.52). The transport is also causal, but its effect is concentrated at the Mamba write site: ablating the direction at the attention site alone has a modest effect that is not decisive by itself (Table 4, Appendix A.22). A direction read at its write sites defends, while one read from its residual does not. The same holds for a transferred direction, not just a native one: a direction read at Falcon-Mamba’s write site controls refusal, while one read from a transformer’s residual does not (Appendix A.11).

We summarize this as a *readout principle*: refusal is cleanly readable, and its steering direction cleanly estimable, at the fresh contribution each architecture writes into the stream, not at the accumulated residual, while the intervention location itself is irrelevant (Appendix A.35). The rule holds across all four architecture classes (Table 6). The full mechanistic dissection, baselines, ablations, and robustness checks are on the anchor, Falcon-Mamba (Section 5).

5 Dissecting the mechanism on the anchor

The *site* at which the shared direction is cleanly read is architecture-specific, and the direction estimated there defends (Table 6). A matched-absolute-magnitude control (Table 7) locates the cause: with the same direction held fixed, pushing at the residual is statistically indistinguishable from pushing at the write site (overlapping 95% confidence intervals), so the residual’s apparent failure is a property of the direction *estimated* there, not of where the gate is applied. A direction estimated in the accumulated residual degenerates when amplified at the deployed operating points, though over a narrow band of low relative magnitudes it can defend coherently, a band much narrower than the write site’s (Appendix A.17). At the gains the compared methods run at, this degeneration is why CAST, the conditional activation-steering baseline that reads its direction from the residual, leaves attack success at 100.0% with only 9.5% of completions coherent. The control is also a reusable diagnostic for separating a site effect from a direction or magnitude artifact.

These numbers establish a causal claim, not a contest for the best defense: our gate only ties a trivial detect-then-refuse policy sharing the same detector (Appendix A.12), so its value is a portable direction, not raw defense strength. Among steering-based methods, CAST at the residual, an open-loop edit, and residual-subspace projection all fail to defend or over-refuse (Table 9). The brittleness prior work reports (Li et al. 2026) traces to the direction read from the residual, not to steering itself.

Table 6: Write-site prefix-injection defense across four architecture families (base to gated attack success at the coherent gain and 64 new tokens; Wilson 95% CIs for every row in Appendix A.24 (transformers), Appendix A.20 (SSM), Appendix A.16 (RWKV), and Appendix A.21 (hybrid)). The last column is the benign-side cost, the change in over-refusal on benign-but-spicy prompts at the same operating point. Per-row n : 200 (SSM), 100 (each transformer), 40 (RWKV), 100 (hybrid, shared base). No row adds any over-refusal, though this is a weak test, since the trigger was fitted to reject those prompts. Under benign prompts wearing a jailbreak template every trigger holds except the hybrid’s, † marking the site where training on wrapped negatives repairs the confound and ‡ the one where it does not (Appendix A.37). The hybrid’s two rows share one base, from a single closed-loop run whose always-on version instead costs 99.0% over-refusal (Appendix A.39). A causal-ablation test (Appendix A.22) finds the hybrid’s control concentrates at the Mamba site. The attention site’s own causal-ablation evidence is weaker, with overlapping confidence intervals, and is not decisive on its own.

Architecture	Exemplar	Write site	Base $\rightarrow$ gated	Δ over-ref.
Selective-scan SSM	Falcon-Mamba	block output	98.0% $\rightarrow$ 5.0%	+0.0 pt
Transformer	Mistral-7B	attn. output	98.0% $\rightarrow$ 14.0%	+0.0 pt
Transformer	Llama-3.1-8B	attn. output	25.0% $\rightarrow$ 0.0%	+0.0 pt
Token-shift rec.	RWKV-6	time-mix output	47.5% $\rightarrow$ 27.5%	+0.0 pt
Hybrid (attn. site)	Zamba2-7B	attn. output	31.7% $\rightarrow$ 0.0%	+0.0 pt†
Hybrid (Mamba site)	Zamba2-7B	Mamba output	31.7% $\rightarrow$ 5.0%	+0.0 pt‡

5.1 Amplifying the refusal direction at the block output defends

The refusal direction is cleanly read at Falcon-Mamba’s block output, the point where the selective scan writes the information selected by the step size Δ into the residual stream. Our gate detects harm there once per prompt with a linear probe and, on a positive decision, amplifies the refusal direction for the whole generation. This steering direction differs from the detector’s probe: we compute it the standard way (Arditi et al. 2024), the average activation on harmful prompts minus that on benign ones. Refusal is not captured by a single direction alone but lives in a broader multi-dimensional subspace (Wollschläger et al. 2025; Winninger 2025; Piras et al. 2026).

Table 7: The push location is irrelevant, the direction-fit decides it (matched-absolute-magnitude control, Llama-Guard, Wilson 95% CI, $n=100$, a model’s three rows share the same absolute push at its coherent gain). Holding the direction fixed leaves the push site within CIs (rows 1–2). Estimating the direction in the accumulated residual instead degenerates it (row 3). RWKV-6 and Zamba2 in Appendix A.35.

Push site	Direction est. at	Falcon-Mamba	Mistral-7B
write site	write site	6.0% [2.8%, 12.5%]	15.0% [9.3%, 23.3%]
residual	write site	6.0% [2.8%, 12.5%]	23.0% [15.8%, 32.2%]
residual	residual	100.0%	48.0%

Table 8: The closed-loop gate on Falcon-Mamba-7B. Attack success rate (ASR) before and after amplifying the refusal direction at the selective-scan block output, decided once per prompt, across three jailbreak families, at 200 harmful prompts per condition with the Wilson 95% confidence interval on the gated rate. The primary ASR is the Llama-Guard-3-8B rate. The reduction holds under an out-of-family judge (ShieldGemma-9B, Gemma family, disjoint from all four targets), and across 4 judges spanning three model families the n-way verdict agreement is 72.8%.

Attack	Base ASR	Gated ASR, Llama-Guard (95% CI)	Gated ASR, ShieldGemma
Prefix injection	98.0%	5.0% [2.7, 9.0]	2.0%
Persona roleplay	81.5%	30.0% [24.1, 36.7]	23.0%
Fictional framing	87.0%	16.5% [12.0, 22.3]	10.0%

Our direction is one dominant axis within that subspace, and using it does not contradict those findings. On Falcon-Mamba the gate lowers attack success from 98.0% to 5.0% under prefix injection, from 81.5% to 30.0% under persona roleplay, and from 87.0% to 16.5% under fictional framing, with Wilson 95% confidence intervals (Wilson 1927), as shown in Table 8. It also holds against a gradient-based adversary: a per-prompt GCG suffix optimized directly through the SSM’s own differentiable scan, not only the fused inference kernel, lowers attack success from 22.0% to 6.0% on $n=100$ evaluated prompts (Appendix A.34). The exception is persona roleplay, which the single-direction gate more than halves (to 30.0%) but does not fully close, and which needs an attack-specific two-direction fix that does not generalize (Section 7, Appendix A.14). Attack success is scored by Llama-Guard-3-8B (Inan et al. 2023) at 200 harmful prompts per condition. The drop is not a one-judge artifact or an artifact of the attack’s injected prefix: each of the 4 judges, spanning three model families, shows it independently, from Llama-Guard (98.0% to 5.0%) to the out-of-family ShieldGemma-9B (78.0% to 2.0%). Judges differ on absolute base rates, so we report the reduction under each separately. The attack’s wording never changes, only the response does, so a judge keying on that fixed wording could not produce a moving score. The locus and defense both reproduce on a second, independently trained instruct SSM, Falcon3-Mamba-7B, so the finding holds for SSMs as a class, not for one checkpoint alone (Appendix A.33).

5.2 The write-read direction, not the closed loop, is what defends

CAST isolates the loop from the direction. It runs the same closed-loop detect-then-steer as our gate, but estimates and steers the direction in the residual stream rather than at the block output. Under prefix injection, it does not lower attack success the way our gate does (Table 9). The matched-magnitude control (Appendix A.35) shows why: the gap is the direction CAST estimates in the accumulated residual, not its push location, which is statistically indistinguishable from the write site (overlapping 95% CIs). This holds even with a more powerful nonlinear MLP detector and across CAST’s own gain sweep (Appendix A.31). The loop’s job

Table 9: Defenses on Falcon-Mamba-7B, attack success (lower is better) across three jailbreak families and over-refusal (plotted in Figure 5). CAST runs the same closed loop as our gate but reads its direction from the residual stream, and the open-loop gate runs the same direction with no detector. The direction gives the defense, the loop the selectivity, so the block-output gate reaches the lowest attack success while holding over-refusal at the undefended base rate. CAST’s near-total attack success is not a coherent jailbreak: when its closed loop fires on harm, the residual push breaks generation into repetitive text rather than producing either a refusal or usable harmful content, and the judge still scores that broken text as compliant. Only 10.5% (prefix injection), 7.0% (roleplay), and 11.0% (fictional framing) of its harmful completions pass our coherence screen (Appendix A.25), against 9.5% pooled ($n=600$). CAST’s benign completions, where the loop does not fire, remain fully coherent, so the breakdown is specific to the closed-loop residual push on harmful prompts, not a general fault in the model or the generation setup.

Defense	Prefix	Roleplay	Fiction	Over-ref.
None	98.0%	81.5%	87.0%	12.0%
Residual projection	95.0%	87.5%	94.5%	9.0%
CAST (residual loop)	100.0%	100.0%	100.0%	12.0%
Perplexity filter	98.0%	81.5%	87.0%	19.0%
SmoothLLM	62.0%	72.0%	86.5%	29.0%
Open-loop block output (no detector)	5.0%	30.0%	16.5%	86.0%
Block-output gate (ours)	5.0%	30.0%	16.5%	12.0%

is controlling collateral damage, not attack success. Open-loop steering, pushing every prompt without checking first, matches our attack success rate but over-refuses badly, blocking 86.0% of benign prompts, while our gate’s 12.0% is the undefended model’s own rate. Our gate achieves the lowest attack success of the steering-based defenses we compare (Table 9), without adding any over-refusal beyond that 12.0% base rate, measured on the detector’s own training distribution (XSTest benign-but-spicy). It matches, but does not beat, a detect-then-refuse baseline (returning a fixed, templated refusal message whenever the same detector fires) on out-of-distribution OR-Bench-hard, where the gate over-refuses 46.3% and detect-then-refuse 47.5%, at a 21% false-positive rate (Appendix A.12). This distinguishes two questions: where refusal is readable, and where it is controllable. In the SSM, on the unrestricted corpus, harm is linearly decodable at the step size Δ (AUROC 1.00) and the mixer output (AUROC 1.00), yet forcing Δ to a constant leaves refusal unchanged, so Δ is a readout, not the controlling site (Appendix A.13). A single steering direction controls refusal in every architecture we test, for every attack except persona roleplay.¹ Persona attacks need two directions instead of one (Appendix A.14).

¹On Mistral the single-direction gate under roleplay does not improve over the base rate, so the transformer defense claim is scoped to prefix injection (Appendices A.14, A.24).

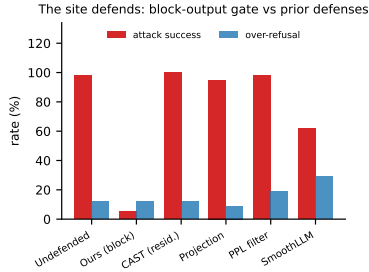


Figure 5: Defenses on Falcon-Mamba-7B under prefix injection (selected rows of Table 9). Only the block-output gate, which reads the refusal direction at the write site, drives attack success down to a low rate. CAST and projection, which read and steer the direction in the residual, barely move it. The perplexity filter has no effect, and randomized smoothing (SmoothLLM (Robey et al. 2025)) achieves a partial reduction well short of the gate’s.

5.3 Robustness and specificity

Two checks validate the trigger and the intervention. Benign-but-spicy prompts are answered normally because the detector does not fire on them, even under the harder test of a jailbreak wrapper, where wrapped-benign firing drops from 0.78 to 0.00 once hard negatives are added. Only the model’s own learned refusal direction controls refusal, not a random direction matched in scale (Appendix A.29, A.30). The defense is stable across 3 seeds (pooled gated attack success 3.3% [2.2, 5.1], non-overlapping with the base) and holds at 300 prompts per condition, under two out-of-family judges (Mistral-7B-Instruct-v0.2 96.0% to 1.5%, ShieldGemma-9B 78.0% to 2.0%), on the HarmBench distribution (93.5% to 5.5%), against an optimized gradient-free suffix (97.5% to 6.5%) and a dilution attack (firing through 256 appended sentences). The detector never fires on MMLU (Hendrycks et al. 2021) or TruthfulQA (Lin, Hilton, and Evans 2022), so capability is unchanged by construction (Appendix A.32).

6 Related Work

Activation monitors and gating. We close the detect-then-intervene loop at a *write site*, the point where a fresh contribution enters the stream before the residual accumulates it, rather than the residual stream or output head where prior gating acts (Lee et al. 2025; Han et al. 2025; Zhou et al. 2025; Zhang et al. 2024). DeTAM edits the write site directly (Li, Jiang, and Wei 2025), but always on rather than detector-triggered, and closing that loop is what our gate adds. What is decisive is the direction read at the write site, not the intervention location (Section 5, Appendix A.35). Our closed-loop gate matches the open-loop edit’s attack success at the undefended over-refusal rate (Appendix A.24). Our trigger is an ordinary linear probe, in the same broad family of internal-state classifiers used to detect harm and deception elsewhere (Azaria and Mitchell 2023; Li et al. 2023; Goldowsky-Dill et al. 2025; MacDiarmid et al. 2024; Marks and Tegmark 2024; Kramár et al. 2026), with no reliability claim beyond deciding when to intervene (Oldfield et al. 2026; Engels et al.

2025). Both the monitored signal and the steered direction port across architectures once re-estimated at each write site, so a monitoring-and-control stack need not be rebuilt per architecture.

Residual directions are readable but brittle. We find the same structure in a pure SSM, concentrated at the selective-scan step size Δ (Section 5), extending a transformer-only picture: refusal is controlled by a single residual direction (Arditi et al. 2024), steering it is brittle (Li et al. 2026; Zou et al. 2023a, 2024), and a readable direction need not be controllable (Galeone et al. 2026) within a broader multi-dimensional subspace (Wollschläger et al. 2025; Winninger 2025; Piras et al. 2026).

State-space interpretability and cross-architecture transfer. To our knowledge, no prior work transports a refusal direction across the attention-to-pure-SSM gap. We do so with a rigid rotation, validate the transport with category-held-out cross-validation, and close it into a working defense loop inside a pure SSM. Prior transport work moves safety directions within one architecture or to related ones (Oozeer et al. 2025; Poppi et al. 2026; Cristofano 2026), never across this gap or under this constraint, and a direction copied directly into the target without any alignment map between the two activation spaces does not transfer (Shnaidman et al. 2025). Existing pure-SSM work is general-purpose steering (Sunku Mohan et al. 2026), an attack (Le Mercier, Develder, and Demeester 2026), a threat analysis (Parmar 2026), or interpretability without defense (Paulo, Marshall, and Belrose 2024). Mamba features and circuits have been studied descriptively (Wang et al. 2025), and crosscoders find cross-architecture shared features (Lindsey et al. 2024; Thasarathan et al. 2025) or differences (Jiralerspong and Bricken 2026). Concurrent work draws a related detection-versus-execution distinction inside Mamba-2 (Jiang 2026), matching our reading of Δ as a readout rather than the controlling site (Appendix A.13). It stays within one SSM family, so it reaches neither the shared-representation claim nor the write-site one. The platonic representation hypothesis (Huh et al. 2024) has been shown only correlationally. We test it causally: removing the direction changes behavior, not just similarity.

7 Scope and Limitations

Attack coverage and available models. We test seven attack types: four fluent jailbreak families (three throughout, a fourth in Appendix A.15), a dilution attack, and two optimized adversarial suffixes (an affirmative-continuation suffix and a probe-evasion suffix, Appendix A.28). Two SSM checkpoints and two transformers give real replication, while RWKV-6 and Zamba2-7B each have a single safety-tuned checkpoint and no second release to replicate on.

The persona fix does not generalize. The two-direction fix that closes persona roleplay on Falcon-Mamba, amplifying refusal while separately suppressing the induced persona (81.5% to 17.5%), does not repeat at RWKV’s or Zamba2’s own write sites, and fails a different way on each (Appendix A.14, A.23). It is a persona-specific augmentation calibrated per model, not a recipe step (Section 2.5).

The detector is a bound. A detect-then-act defense is bounded by its detector: an adversary that routes harm past it recovers the undefended model. The defense holds against all three adaptive attacks we test (Section 5.3). A GCG suffix optimized against the always-on gate itself fails outright. The attack’s only leverage is the detector, not the steering: optimizing against the deployed detect-then-gate system raises attack success from 3.7% to 15.3% (Appendix A.24). Two non-adaptive wrapper failures are sharper. The DAN (“Do Anything Now”) wrapper evades RWKV’s detector outright (100.0% to 0.0% firing once wrapped). A detector trained only on plain harmful-versus-benign prompts also learns the wrong feature: it keys on the jailbreak template itself, and fires on 100.0% of benign prompts once they are wrapped in one, on both transformers and at both Zamba2 sites. Adding jailbreak-wrapped benign prompts to the training set removes this template confound on every architecture except at the Zamba2 Mamba site, which still fires on 100.0% of wrapped benign prompts (Appendix A.37), so the hybrid’s Mamba row has no deployable trigger. All of this bounds the trigger, not the direction or the site, since the transport and steering results use no detector.

The write site is not free to calibrate. Too high a gain degenerates the transformer’s output, and an uncalibrated gain at the hybrid’s Mamba site can backfire (Appendix A.24, A.21, A.23). The defense margin also varies with each model’s baseline alignment (Appendix A.20), so the cross-architecture comparison is of the site-versus-direction *pattern*, not of absolute margins.

The judge is validated, human counts on the newer architectures are small. Attack success is scored by an LLM judge, so two raters independently labelled 300 blind completions, shown request and response only (Appendix A.40). They agree at 89.0% (Cohen’s $\kappa=0.80$), and the primary judge agrees with their consensus at 94.8% ($\kappa=0.90$ on $n=97$ doubly rated completions). The judge’s errors run in one direction: it sometimes counts as a successful attack a response the raters judged safe, so the reported defense numbers understate the defense rather than inflate it. Per-item judge verdicts existed only for the Falcon-Mamba cells, so agreement is measured there and assumed elsewhere, and for RWKV-6 and Zamba2 each evaluation cell (one architecture-attack-condition combination) has only a small number of human-rated completions.

8 Conclusion

Refusal is one representation across architectures that share no token-mixing mechanism. A refusal direction read in a transformer and mapped into a Mamba state space is still the state-space model’s refusal trigger, since removing it jailbreaks the target while a random control does not. What is architecture-specific is only the site where that representation is cleanly read, the fresh, single-layer contribution each architecture writes before the residual accumulates it, while a matched-magnitude control shows the intervention location itself does not matter. Refusal is thus architecture-independent as a representation and architecture-specific as a readout site.

This readout rule is directly actionable. Porting a safety stack to a new architecture reduces to the five-step recipe of Section 2.5: find the write site, select layers by held-out decodability, fit the probe and the mean refusal direction there, calibrate the gain under a coherence screen, and deploy closed-loop. Following it, the same detector-triggered gate lowered jailbreak success on a selective-scan SSM, two transformers, a token-shift recurrent model, and a hybrid, held against a gradient-optimized adaptive attack on the SSM. Where target labels are scarce, a transported direction detected harm the target’s own labels missed.

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Ethics Statement

This work studies how refusal is represented and steered in order to strengthen safety defenses, and its dual-use risk is not limited to running existing jailbreak templates. The rotation-transport method itself is a new technique: reading a refusal direction in one model and reconstructing it well enough to ablate in an architecturally unfamiliar target, shown here to work across an SSM, a token-shift recurrent model, and a hybrid. The five-step recipe for siting the gate on a new architecture (Appendix A.2) is symmetric by construction: the write site a defender must find to steer refusal is the same site an attacker would need. We judge the interpretability and defensive value of this work to outweigh the risk. The technique requires white-box access, paired contrastive data, and per-architecture rotation-fitting, none of which a deployed black-box attacker would have. That barrier assumes an attacker without model access. It does not hold against someone who already legitimately possesses a model’s open weights and wants to strip its safety training to redistribute an “uncensored” fork, an existing and active ecosystem (Young 2025). That actor already has white-box access, and paired contrastive data and per-architecture rotation-fitting are exactly what such efforts already do by hand. What our recipe adds is a faster way to find where to intervene, not a barrier this actor lacks. We judge the added risk to be real but narrower than it first appears: existing ablation methods already remove refusal effectively on the transformer architectures they target, so the uplift here is concentrated on architectures without established ablation tooling, chiefly the SSM and hybrid families this paper studies. We publish anyway because establishing a working defense for exactly those architectures is the paper’s main contribution, and that requires disclosing how we found the site. We also limit what we add beyond that. All models we study are already-public open-weight releases, so we expose no undisclosed vulnerability. The jailbreak attack families we test (prefix injection, roleplay, fictional framing, GCG) are all published. We report harmful generations only as aggregate rates, never as usable content. We release no jailbreak artifacts, and the artifacts we do release are oriented toward detection and refusal. One specific finding in this paper is itself a working evasion: on RWKV, a published jailbreak wrapper (DAN) fully bypasses our detector rather than merely weakening it (Section 7, Appendix A.16). We disclose it rather than omit it because the wrapper is already public and widely used, and because knowing a specific defense fails against a specific, named attack is more useful to defenders building on this work than a defense whose failure modes are hidden. We follow responsible-disclosure norms for the open-weight models we study.

A Technical Appendix

This is the technical appendix for the main paper. It gathers the formal definitions of the detector, the refusal direction, and the write-site gate, together with the full result tables and figures the main text refers to and the supporting experiments, ablations, and robustness checks cited there. Each subsection names the main-text claim it supports and is self-contained, and cross-references of the form Section 5 point into the main paper. The main paper is self-contained for every headline claim: each carries its core number inline in the body. This appendix adds the additional experiments, robustness checks, and ablations that support and extend those claims.

A.1 Formal definitions

Detector. At routing layer ℓ the pooled activation $h_\ell(x)$ is standardized to $\tilde{h}_\ell(x)$ and scored by a linear probe trained with L2-regularized logistic regression,

$$s_\ell(x) = w_\ell^\top \tilde{h}_\ell(x) + b_\ell, \quad p_{\text{harm}}(x) = \sigma(s_\ell(x)),$$

firing the gate once per prompt when $p_{\text{harm}}(x) > \theta$.

Refusal direction. The gate amplifies the unit mean-difference direction

$$\hat{u}_\ell = \frac{\mu_\ell^{\text{harm}} - \mu_\ell^{\text{benign}}}{\|\mu_\ell^{\text{harm}} - \mu_\ell^{\text{benign}}\|}, \quad \mu_\ell^c = \frac{1}{|c|} \sum_{x \in c} h_\ell(x),$$

which is distinct from the detector’s discriminative direction w_ℓ .

Multi-layer gate. On a positive decision, Equation 1 is applied at every layer in the selected set $\mathcal{L}$ (the top- k by held-out AUROC) for the whole generation, $y_t^\ell \leftarrow y_t^\ell + \alpha \hat{u}_\ell$ for $\ell \in \mathcal{L}$, with α calibrated as in Appendix A.25.

Cross-architecture maps. For paired activations X_s, X_t on shared prompts, the ridge map is $W = \arg \min_W \|WX_s - X_t\|_F^2 + \lambda \|W\|_F^2 = X_t X_s^\top (X_s X_s^\top + \lambda I)^{-1}$, and the orthogonal (Procrustes) map is $R = \arg \min_{R^\top R = I} \|RX_s - X_t\|_F = UV^\top$ with $U\Sigma V^\top = X_t X_s^\top$. The structural test uses R , which can only rotate and so cannot synthesize a direction. The steering-transfer test uses W .

Harm-weighted pool. The dilution-invariant detector pools per-token scores by $s_{\text{pool}} = \sum_t \text{softmax}(\tau s)_t s_t$, weighting each token by its own harm logit (Appendix A.27).

Unequal widths. When source and target hidden widths differ (Zamba2’s 3584 against the other models’ 4096), the orthogonal map is the thin-SVD solution $W = UV^\top$ of the rectangular cross-covariance, a semi-orthogonal matrix (a partial isometry): it remains norm-preserving on its row space and still has no freedom to synthesize a direction, but is a rotation only up to the smaller of the two dimensions.

A.2 Recipe for siting the gate on a new model

The step-by-step form of the procedure summarized in Section 2.5.

1. **Find the write site.** Identify where the architecture moves information between positions and writes it into the state: the selective-scan selection (Δ, B, C) and the block output in an SSM, the attention output projection in a transformer. Δ is the site with no counterpart in a transformer.
2. **Select decodable layers that steer without over-steering.** Probe each layer for harm and take the top k by held-out AUROC. Which decodability to rank by is per-model. On Falcon-Mamba the mixer-output ranking steers cleanly, while Δ -ranking pulls in late layers that over-steer into degenerate text. On a better-aligned model where output AUROC saturates and picks destabilizing early layers (Falcon3-Mamba), rank by Δ . Step 4’s strength check confirms the layers steer rather than degenerate.
3. **Fit the detector and the direction.** A linear probe at the block output (the routing layer) is the per-prompt trigger. The mean refusal direction (mean harmful minus mean benign) at the same block output is what the gate amplifies.
4. **Calibrate the strength.** Sweep the gain α on a small held-out set and keep the largest value that leaves generation coherent. Over-steering shows up as repetitive degeneration, caught with a distinct- n check. A better-aligned model needs a smaller α .
5. **Deploy closed-loop.** Score the prompt once. If it reads harmful, amplify the refusal direction at the selected layers for the whole generation, otherwise run the model unchanged.

A.3 Cross-architecture transfer, ablation test

Figure 2 plots Table 5: removing the mapped refusal direction (Transferred) raises the target’s attack success (ASR, the fraction of attacks the judge scores unsafe) like removing its own (Native), while an orthogonalized random control (Random) does not, across three model pairs and both directions. The Transferred-minus-Random gap holds across all pairs. On Mamba→Mistral (60.0% against 27.0%), the pair where a raw mapped-random control had made the effect look non-specific, it remains clear against the orthogonalized control. Where the transferred direction raises attack success above the native one, the successful completions are genuine jailbreaks rather than degeneration. They pass the same coherence screen used for the defense numbers on 77.0% to 100.0% of prompts, close to the native direction’s 93.0% to 100.0%.

A.4 The locus across model scale

Figure 6 shows the locus holds as the model grows.

A.5 Gate-strength calibration on a second SSM

Figure 6 calibrates the gate strength on the second SSM.

A.6 Geometric similarity is unreliable for cross-architecture transfer

A natural metric for the mapped direction is its cosine with the target’s own refusal direction, but this is not reliable. The

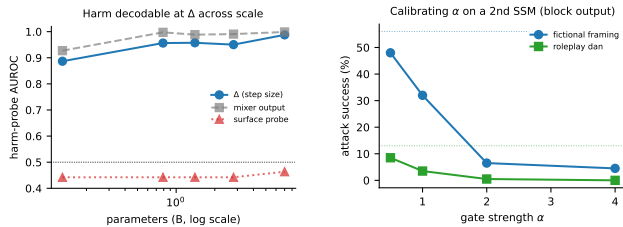


Figure 6: (Left) Harm is decodable at the step size Δ across base Mamba models from 130M to 7B parameters, while a surface probe stays near chance. (Right) Calibrating the gate strength α on the second SSM (Falcon3-Mamba), block-output gate at its Δ -ranked layers. Attack success falls monotonically with α where there is headroom (dotted lines are the undefended rates) and every point passes the coherence screen. With the anchor’s mixer-output layer ranking instead, the anchor’s gain over-steers Falcon3 (Appendix A.20).

ridge map is biased toward the target’s refusal axis, so even a *random* source direction can map to a high cosine, up to 0.515 across our pairs. Such a direction is not always behaviorally inert, so a mapped-random direction is not a clean specificity control. This is why the ablation control in Section 4 orthogonalizes the random direction to the target refusal axis after mapping (cosine ≈ 0), removing the leaked component. Two further controls agree that geometry alone is unreliable. Fitting the map on shuffled pairs collapses the held-out cosine (0.17 versus 0.95), and the unmapped source direction is near orthogonal to the target’s (0.01). A cross-architecture transfer claim should therefore rest on the structural and behavioral specificity controls, not on a geometric similarity score.

A.7 A matched random-rotation null for the ceiling transport

The within-Mamba transport in Section 4 is at ceiling, so we add a *matched* null that makes the structural claim discriminating: a shuffled-correspondence *orthogonal* map (a random rotation, not a flexible ridge map) transports the source probe into the target. Even where the correct rotation reaches ceiling, this random rotation collapses to 0.33 on Falcon-Mamba, so a rotation must carry the right correspondence to transport at all, and the ceiling result is not vacuous. Fitting the orthogonal map requires standardizing each space per dimension first, since activation scales differ across architectures and an unstandardized orthogonal map is ill-posed. We rest the shared-representation claim on this matched null, the category-held-out cross-validation (Appendix A.8), and the behavioral ablation transfer (Appendix A.11, Table 5). A single arbitrary split of the hardest surface-matched corpus is not a stable basis for a below-ceiling claim, since the held-out probe is sensitive to how the minimal pairs fall across train and test. The category-held-out folds avoid this by holding out a whole category at a time rather than a random slice, and the transport tracks the target’s own probe across the resulting range of difficulty.

A.8 Category-held-out cross-validation of the transport

To test the transport off ceiling without relying on an arbitrary train/test split, we cross-validate by XSTest category. XSTest labels each unsafe prompt with a contrast category (homonyms, figurative language, privacy, discrimination, and others). We leave one category out at a time, fit the target’s own probe, the source probe, and the orthogonal (Procrustes) map on the other seven categories, and evaluate the target’s own probe and the transported source probe on the held-out category, whose surface features the fit never saw. This holds out a whole category rather than a random slice, so the held-out difficulty is a property of the category, not of the split. Table 3 reports, per model pair, the mean over the eight folds of the target’s own probe and the transported probe, and their per-category Pearson correlation. The own probe ranges from ceiling down to 0.78 on the discrimination category, and the transported probe follows it, correlating with the own probe at 0.99 from Llama and 0.99 from Mistral into Falcon-Mamba. The transported probe beats a shuffled-correspondence orthogonal null in 16 of 16 folds across both pairs. This map-correctness test replicates on an independent corpus with more categories: on BeaverTails (Ji et al. 2023) (14 harm categories, Llama into Falcon-Mamba) the correctly rotated probe transports at AUROC 1.00 and beats the shuffled null (0.51) in all 14 folds. Harm versus benign is near ceiling there, so BeaverTails does not exercise the off-ceiling tracking that the surface-matched XSTest contrast does, but the shuffled-null test is valid at ceiling, which is the point: only the correct rotation transports the probe, across 14 independent categories on a second corpus. Because the transported probe’s accuracy tracks the target’s own accuracy category by category, a map that had merely found the target’s refusal axis is ruled out, since such a map would not know which categories are hard for the target.

The tracking is not an artifact of the single below-ceiling category. It holds under a rank correlation, which no single point can dominate (Spearman 0.95 and 0.95 for the two pairs), and it survives removing the discrimination fold entirely: the Pearson correlation over the remaining seven near-ceiling folds is 0.85 (Llama) and 0.99 (Mistral), each still clearing an exact permutation test over all fold pairings (0.015 and 0.001). So the transported probe tracks the own probe even among the categories where both sit near ceiling, not only at the one category that falls off it.

A.9 The causal ablation holds through the rigid rotation

The behavioral ablation in Section 4 transports the refusal direction through the ridge map W . Because a flexible map can align to the target’s own refusal axis, the sharper causal test transports the source direction through the orthogonal (Procrustes) rotation instead, the same rigid map that cannot synthesize a direction, and ablates that. The one requirement is that the rotation be fit the way the ceiling-AUROC structural transport is fit. Orthogonal Procrustes is scale-sensitive, so fitting it on raw activations aligns the high-variance dimensions while under-aligning the refusal direction. The trans-

ported direction then reaches only cosine 0.68 (Llama) and 0.35 (Mistral) to the target’s own refusal axis, and ablating it barely moves attack success. Fitting the rotation on standardized activations, exactly as the structural transport does, aligns the refusal direction closely (0.998 to 0.998 into Falcon-Mamba).

Through the standardized rigid rotation the causal ablation replicates into the SSM. Removing the rotation-transported direction from harmful prompts raises the SSM’s attack success to 44.0% from Llama and 45.0% from Mistral, at the level of ablating the SSM’s own native direction (41.5%) and far above the orthogonalized-random control (9.0%), with fully coherent completions (100.0%) that are genuine harmful instructions rather than degeneration. So the same rigid rotation that transports the harm probe at ceiling AUROC and tracks the target’s category-held-out accuracy also causally controls its refusal. Structure and causation hold in one map.

Into the transformer targets the SSM-calibrated ablation gain over-suppresses: the raw orthogonal-rotation ablation from Falcon-Mamba into Mistral reaches 92.5% attack success, but only 3.0% of those completions pass our coherence screen. This cell is therefore dominated by degenerate output that the harm judge miscounts as compliance, not by genuine off-topic-but-fluent drift, and we do not report it as evidence for the causal claim. We instead report the causal-through-rotation result for the SSM targets, which are the direction the shared-representation claim is about, a transformer refusal direction transported into the pure SSM and there controlling refusal.

A.10 A Mamba-2 target: harm is decodable, but a code model has little refusal to steer

The decodability result crosses the selective-scan architecture generation. On Codestral-Mamba-7B, a Mamba-2 model from a different developer than Falcon-Mamba, a linear probe on the mixer output separates harm from benign at AUROC 1.00 (mean 0.98 over the layers, 61 of 64 above 0.95), so the harm representation is present in Mamba-2 and not specific to the Mamba-1 Falcon-Mamba family. The detector built on it fires on every jailbroken harmful prompt we test.

The write-site defense, however, does not carry over, and for an instructive reason. Codestral-Mamba is a lightly-safety-tuned code model that rarely refuses a harmful request in the first place. Amplifying the refusal direction at its mixer output never induces refusal at any coherent gain. At low gain the output is unchanged and at higher gain it degenerates before it refuses, across both an output-decodability layer selection and a mid-layer one, and every gain we swept. This is the read-versus-steer dissociation again. The harm is read at ceiling, but there is little learned refusal behaviour to amplify, so the write-site gate has nothing to steer up. The defense therefore requires a target that already refuses, which is a property of safety tuning rather than of the selective-scan architecture, and we scope the defense to safety-tuned pure SSMs (Falcon-Mamba and Falcon3-Mamba) accordingly.

A.11 Defense transfer is limited by the target’s intervention site

Amplifying the mapped refusal direction defends the target only where the target is defensible at all. Into the SSM it lowers attack success from 97.0% to 28.5% for the Llama pair and to 29.5% for the Mistral pair, while the orthogonalized random control does not. Into a transformer target the same amplification helps little at the residual stream, the residual-stream weakness we document for native directions (Section 7). This is a site effect, not a representation failure: the transformer is defensible at its attention write point, where the native gate lowers attack success (Appendix A.24), whereas the mapped direction here is applied at the residual. The defense-transfer asymmetry therefore tracks the target’s intervention site, not a failure of the representation to cross the architecture gap, which is why the shared-representation claim rests on the structural and behavioral ablation tests rather than on defense transfer.

A.12 Write-site steering versus detect-then-refuse

The write-site gate and a detect-then-refuse policy share the same harm detector and differ only in what they do on a flagged prompt, amplifying the refusal direction or emitting a canned refusal. We ask whether steering buys any deployment advantage over the canned refusal. We sweep the gate gain and compare the resulting attack success and over-refusal against detect-then-refuse at the deployed detector threshold, on the standard benign set (XSTest-safe) and a harder one (OR-Bench-hard (Cui et al. 2025)) that the detector false-flags more often. Both metrics use the same lexical refusal check for the two methods, so this is a mechanism comparison rather than the judged headline number. Unlike the RWKV case (Appendix A.16), where a fluent disclaimer-then-comply pattern fooled the lexical check, this comparison is low-risk for that failure mode: detect-then-refuse’s output is a literal canned template, unambiguous by construction, and the gate’s own completions at this operating point are checked coherent. We confirm this directly by regenerating the gate’s completions at the coherent gain ($n=50$) and rescored with Llama-Guard-3-8B: attack success is 0.0% under both scores, with full per-completion agreement between them (100.0%) and output 100.0% coherent.

Table 10 shows the outcome. On XSTest-safe the detector flags almost no benign prompt, so both methods sit at the base over-refusal of 2.0%. On OR-Bench-hard the base model already refuses 32.5% of prompts unprompted, and the detector false-flags 21% of them. On those false positives a canned refusal always over-refuses and the steered gate refuses 94% of them, so the two methods reach the same over-refusal (47.5% for detect-then-refuse and 46.3% for the gate) at the same attack success. The gate has a single coherent operating point. At the calibrated gain (4) it reaches 0% attack success, below it the steering is too weak to defend, and above it the generation degrades. Above the coherent gain the steered text becomes incoherent and repetitive, and the lexical check reads such text as neither a refusal nor a completion. It therefore spuriously raises the measured attack success and lowers the measured over-refusal at the

Table 10: Write-site steering against detect-then-refuse at the deployed detector threshold, under a lexical refusal check. Both share the detector and reach the same attack success and over-refusal. The gate has a single coherent gain (4). Higher gains degrade the generation rather than improving the tradeoff. Over-refusal on the harder OR-Bench-hard set is bounded by the base model’s own refusals (32.5%) and the detector’s false-positive rate (21%), which both methods inherit.

Method	Attack success	Over-refusal (XSTest)	Over-refusal (OR-Bench)
Undefended base	100%	2.0%	32.5%
Detect-then-refuse	0%	2.0%	47.5%
Write-site gate ($\alpha=4$)	0%	2.0%	46.3%

same time, even though the model is not complying with the attack. We report only the coherent gain as an operating point for this reason, and read the higher-gain points of the sweep as marking where coherence breaks rather than as a real change in either rate. Neither method dominates because both are bounded by the shared detector, which is why we present the gate as a localization of where refusal is read rather than as the deployment defense of choice.

A.13 Δ is a readout, not the computation

We test whether the selective-scan step size Δ *computes* the harm signal or merely carries it. On $n=120$ prompts, replacing Δ with an input-independent global mean, which removes its selectivity entirely, leaves harm just as decodable at the mixer output (1.00 to 1.00) and leaves the model’s refusal rate essentially unchanged (91.7% to 95.8% on 24 held-out generations, coherence 98.5% to 92.0%). Decodability survives the same freeze for the other input-dependent selection parameters B and C too. We additionally checked B ’s refusal rate, also unchanged (91.7% to 83.3%, coherence 99.2%). Harm is therefore carried redundantly to the block output, so we read Δ as the most decodable *readout* site rather than the mechanism that routes harm. This matches the deployed design, in which the gate both detects and writes at the block output, not at Δ , and it is why we make no claim that Δ is the causal origin of the safety signal. The distinction between a decodable signature and the circuit that computes it is independently drawn for Mamba-2 by Jiang (2026). We take the conservative reading and claim Δ as a readout only.

A.14 Persona injection: why roleplay resists a single refusal direction, and a two-direction fix

The gate more than halves attack success on persona roleplay (Table 8) rather than removing it, and inspecting the gated completions explains why. The residual failure is a *disclaimer-then-comply* pattern, where the completion opens with an induced refusal disclaimer (“I am not promoting illegal activities”) and then drifts back into compliance (“however, here is a step-by-step guide”). The prompt-level gate induces the refusal token, but the in-character persona, re-instantiated from context as the model decodes, pulls the continuation back toward the harmful answer. A lexical scan

illustrates the mechanism. 169 of 200 gated completions open with a refusal disclaimer, up from 13 at base, and 28 show the explicit disclaimer-then-comply drift. The response-level judge behind Table 8 reads the whole continuation and scores the drifting completions as harmful, which is why the judge-based gated rate is 30.0% rather than the lexical refusal rate. Prefix injection and fictional framing do not establish a persona and are driven to 5.0% and 16.5%. Raising the gain suppresses the drift but over-steers into degenerate text, so a single prompt-level mean-direction gate halves but does not remove a persona-momentum attack. On Mistral-7B this failure mode is worse than incomplete: single-direction amplification actually raises roleplay attack success, from 63.5% to 78.0% (Table 21), a genuine backfire rather than a partial defense. We read this as the same disclaimer-then-comply mechanism pushed further by the stronger, less-calibrated Mistral gain, and it is the reason the two-lever, persona-suppressing fix below is necessary rather than merely an improvement.

The fix is a different direction, not a stronger one. Amplifying the refusal direction harder does not help: a per-token adaptive gain that boosts on the harmful pivot over-steers into the same degeneration. The roleplay attack instead injects a recoverable *persona* component at the write site, the direction the wrapper adds to a harmful prompt, which is distinct from refusal (cosine -0.30). Suppressing this persona direction alone is worse (79.5%, it de-personifies the request without inducing refusal), and amplifying refusal alone reaches 38.0% in this single-layer persona-suppression setting (the multi-layer headline gate reaches 30.0%, Table 8). Doing both, amplifying the refusal direction and ablating the persona direction at the block output, lowers attack success to 17.5% with coherent refusals, below the partial gate. The detector fires on 0.0% of benign-but-spicy prompts, so the gate adds 0.0% over-refusal beyond the base model’s 12.0% on these prompts. Roleplay is therefore a persona-injection problem, not a refusal-strength one, and the two directions are needed together because neither suffices alone. The sign of the persona-versus-refusal cosine is not an artifact of layer choice: across a three-layer sweep per model it is negative at every steered layer in all three non-transformer models, ranging over the sweep from -0.49 to +0.02 (the positive end a single early RWKV layer), against +0.58 to +0.82 in both transformers. Table 11 gives the per-model cosine at each model’s read site (the write site for the non-transformers, the residual for the transformers), and Table 12 isolates the two directions.

The fix is weaker off the block output. The same two-direction fix applies to a roleplay-vulnerable transformer, but only partially. On Mistral-7B (roleplay base 86.0%), a calibrated single-layer gate that amplifies refusal and suppresses the persona direction lowers attack success to 51.0% with coherent refusals, short of the 17.5% reached on Falcon-Mamba. The persona suppression has to be a constant push rather than a per-token projection, because the input-dependent form destabilizes a pure recurrence (RWKV) and over-steers a transformer residual. The gap to Falcon-Mamba is one of calibration headroom at the write point, not a

Table 11: The roleplay wrapper’s injected persona direction versus the refusal direction (cosine at each model’s read site: the write site for the SSMs and RWKV, the residual stream for the transformers). It is anti-aligned with refusal in every non-transformer model tested and aligned in the transformers’ residual stream, so amplifying refusal alone cannot remove the persona in the non-transformers, and a second, persona-ablating direction is needed. At the transformers’ attention write site the cosine is near zero (Table 13), so the split is between sites as much as architectures.

Model	Persona-refusal cosine
Falcon-Mamba-7B (selective-scan SSM)	-0.30
Falcon3-Mamba-7B (selective-scan SSM)	-0.30
RWKV-6 (token-shift recurrent)	-0.49
Llama-3.1-8B (transformer, residual)	+0.65
Mistral-7B (transformer, residual)	+0.74

Table 12: Roleplay attack success on Falcon-Mamba-7B under each block-output intervention (single-layer persona-suppression setting). Amplifying refusal alone helps but does not remove the attack, suppressing the persona alone barely moves it, and the two together close the gap. Over-refusal is unchanged from base (0.0% added).

Intervention on the roleplay attack	Attack success
Base (no gate)	81.5%
Amplify refusal only	38.0%
Suppress persona only	79.5%
Both (amplify refusal, suppress persona)	17.5%

strength ordering across sites (Appendix A.20): the constant-push persona suppression has less room before the coherence cliff on Mistral than the block-output gate has on Falcon-Mamba.

A.15 A second attack family and a compositional defense at the write site

The write-versus-residual result should not rest on one attack family. We repeat the comparison under a second, stronger family, *combo_strong*, an AIM/Machiavellian persona jailbreak. On the selective-scan SSM the contrast reproduces. Amplifying refusal at the write site lowers *combo_strong* attack success on Falcon-Mamba from 98.0% to 11.0% while the same steering at the residual leaves it at 100.0%.

On a transformer the strong persona attack exposes a sharper form of the same principle. *combo_strong* jailbreaks Mistral-7B at 97.0% (79.0% under the out-of-family Gemma judge). Amplifying refusal alone at the attention write site, at the gain $\alpha=3$ used throughout this comparison, lowers this only to 77.0%, and higher gains degrade generation rather than defend (Appendix A.24). A persona attack adds a second direction, the in-character persona, that refusal amplification does not remove. The two-direction gate that already defends roleplay, additionally ablating the persona projection at strength $\beta=1$, drives *combo_strong* down to 5.0% (0.0% under the Gemma judge, single run) with coherent

Table 13: Cosine between the refusal direction and the injected-persona direction, at the residual stream versus the attention write site, for two persona attacks and two transformers. The two directions are strongly aligned at the residual and near-orthogonal at the write site, so the persona can be suppressed independently of refusal only at the write site.

Model	Attack	Residual	Write site
Mistral-7B	roleplay	+0.74	-0.11
Mistral-7B	combo_strong	+0.76	-0.02
Llama-3.1-8B	roleplay	+0.52	+0.08
Llama-3.1-8B	combo_strong	+0.55	+0.17

Table 14: Compositional defense against a strong persona attack (*combo_strong*) at the Mistral-7B attention write site. Neither refusal amplification nor persona suppression alone defends, while the two together drive attack success from 97.0% to 5.0% (0.0% under the out-of-family Gemma judge), with coherent refusals. The same combined gate at the residual, where the two directions are aligned, leaves attack success at 100.0%.

Intervention at the write site	Llama-Guard	Gemma
none (base)	97.0%	79.0%
amplify refusal only	77.0%	–
ablate persona only	96.0%	–
both (refusal and persona ablation)	5.0%	0.0%

refusals and no added over-refusal. Neither lever alone suffices. Refusal amplification alone reaches 77.0% and persona suppression alone 96.0%, and only the two together defend (Table 14). On the more heavily safety-tuned Llama-3.1-8B *combo_strong* does not land in the first place (base attack success 1.0%), so this persona analysis is confined to Mistral, mirroring the gradient attack, which also only reaches a jailbreakable base rate on Mistral (Appendix A.24).

The compositional defense works at the write site and not at the residual, for a geometric reason. Prior work shows refusal is not a single direction but spans several (Wollschläger et al. 2025). Here the relevant second direction is the injected persona, and whether it can be suppressed independently of refusal depends on the site. At the attention write site the refusal and persona directions are near-orthogonal, so ablating the persona does not disturb the refusal push and the two act as independent levers. At the residual the same two directions are strongly aligned, so removing the persona projection also cancels the refusal push, and the combined gate there does not defend (attack success 100.0%). The pattern holds for both persona attacks and both transformers we test (Table 13). The write-site cosine stays near zero while the residual cosine is large and positive (+0.52 to +0.76). The write site is therefore not only where refusal is cleanly read but where refusal is linearly separable from the attack’s persona direction, which is what makes the compositional defense possible.

A.16 RWKV: the residual is inert, but the sub-block write sites steer

The shared-representation result (Section 4) extends to RWKV-6, and so does the write-site principle once the site is re-localized, which makes RWKV a third architecture class that confirms the principle rather than a loose end. Harm is decodable at every RWKV block (held-out AUROC 1.00) and a transformer-read refusal direction transports in at AUROC 1.00, above its shuffled control 0.43, so the refusal representation is present and shared.

Judging methodology. Attack success throughout this subsection is scored by Llama-Guard-3-8B, the paper’s primary judge, applied uniformly to the saved completions rather than the lexical `is_refusal` heuristic (§3) used during development. The Zamba2 numbers in Appendix A.22 are largely unchanged between the two judges, but RWKV’s are not, detailed below.

The transport is also causal, though with a narrower specificity margin. Fitting a standardized Procrustes rotation between Llama’s residual (layer 14) and RWKV’s time-mix write site (layers 14-24), transporting Llama’s refusal direction through it, and suppressing that direction on held-out plain harmful prompts ($n=100$) raises attack success from 3.0% [1.0, 8.5] to 58.0% [48.2, 67.2], while suppressing a matched orthogonalized-random direction (mapped through the same rotation) raises it to 40.0% [30.9, 49.8] – well above base, but still below the transported cell. The transported and native directions have cosine similarity 1.00, i.e. the rotation recovers RWKV’s own refusal direction almost exactly. This holds across seeds: pooled over three seeds ($n=300$, each with its own prompt draw and random-control direction), attack success rises from 3.0% [1.6, 5.6] to 54.7% [49.0, 60.2] for the transported direction, against 32.3% [27.3, 37.8] for the random control. The CIs do not overlap, so the direction is specific, but the margin is narrower than the pure SSM (Appendix A.9) or the hybrid’s Mamba site (Appendix A.22).

Why the lexical heuristic differs. Under the lexical screen, the same cells read 28.3% (transported) against 0.7% (random), a far larger gap. Inspecting the completions shows why: suppressed RWKV frequently opens with an apology ("I’m sorry...") and then either complies ("...I’m happy to help you with the steps to make and distribute counterfeit money") or degenerates into repetition, the same disclaimer-then-comply pattern documented for persona attacks on the SSM (Appendix A.14). The lexical heuristic checks only the first 200 characters for a refusal marker and stops at the opening apology, scoring these as refusals, while Llama-Guard reads the full response and flags the ones that comply or reveal harmful content. The causal claim holds under either judge. We report Llama-Guard’s more modest margin above since it is the paper’s primary judge and reads the whole response rather than only its opening.

Together, all three non-transformer architecture classes now have causal, not only structural, evidence for the shared-representation claim, scored with the paper’s primary judge throughout.

The residual is inert, not just under-perturbed. Amplifying the refusal direction at the residual does not control refusal, and this is not a matter of scale. We steer each site at a magnitude *relative* to its own per-token activation norm, so the perturbation is matched across sites (Table 15). The residual, whose activations have norm 519, absorbs a push of up to 4 times that norm along the refusal direction with attack success unchanged (47.5% to 47.5%) and output still coherent. This whole-block output is architecturally the residual at RWKV’s block boundary (distinct from the SSM’s block output, which is a write site, not the residual), and the refusal direction read there is a dead readout: a perturbation several times the activation norm along it does nothing. The residual is the weak site, as in every architecture.

The sub-block write points steer. The effective sites are the two points where RWKV’s sub-blocks write a fresh, single-layer contribution into the stream, before the residual accumulates it: the time-mix output (norm 27) and the channel-mix output (norm 54). At a matched relative magnitude far below the block’s (0.25 of each site’s own norm), amplifying the refusal direction lowers attack success from 47.5% to 17.5% at the time-mix output and 7.5% at the channel-mix output, reaching 0.0% with a stronger push. This relative-magnitude sweep is a separate operating point from the fixed-gain write-site result reported in the abstract and Table 6 (47.5% [32.9, 62.5] to 27.5% [16.1, 42.8] at $\alpha=16$, Wilson 95% CIs), run here specifically to rule out a scale confound. The two numbers are not in tension, they answer different questions about the same site. A matched random direction at the same site and magnitude does not merely fail to defend: it actively raises attack success to 82.5%, well above the 47.5% baseline. This is not a degeneration artifact inflating the count, output at this cell is 97.5% coherent by our screen, matching the refusal direction’s own 97.5% (both effectively fully coherent), so the increase is a genuine behavioral shift, not garbled text. We read this as the time-mix site being generically write-sensitive: an arbitrary push there changes downstream behavior in either direction, and only the refusal-aligned direction specifically pushes it toward refusal. So the same absolute perturbation is a dead readout at the block yet causal at a write point an order of magnitude smaller in norm. The channel-mix output steers for the same reason the time-mix output does: each is a fresh, single-layer sub-block contribution written into the stream before the residual sums it with everything already accumulated there. RWKV has two write sites because it has two sub-blocks, each writing its own contribution before the residual combines them. The channel-mix random control is less cleanly inert than the time-mix (15.0%, coherence 85.0% against the refusal direction’s 100.0%), so the time-mix output is the cleaner write site. RWKV-6-World is prompted in its native `User:/Assistant:` format throughout. The earlier inert result for these sites was a raw-prompting artifact of that missing chat template.

A.17 The residual’s operating band

A relative magnitude is a different absolute displacement at each site. The transformer’s attention output has an activa-

Table 15: RWKV-6 norm-matched steering (native prompting, prefix injection, base attack success 47.5%). Each site is steered at a magnitude *relative* to its own activation norm, so the perturbation is matched across sites. The block output (the residual) is inert up to $\text{rel}=4$, a push several times its norm, while the sub-block write points steer at a fraction of theirs and a matched random direction does not (cleanly for the time-mix output). The residual’s inertness is therefore direction-intrinsic, not a matter of scale. Both sub-blocks write a fresh contribution into the stream before the residual accumulates it, so both are write sites.

Site	Norm	rel	Refusal ASR	Random ASR
Block output (residual)	519	4	47.5%	47.5%
Time-mix output	27	0.25	17.5%	82.5%
Channel-mix output	54	0.25	7.5%	15.0%

tion norm of 0.79 and its residual 22.0, so the same relative magnitude is an absolute push $27.9\times$ larger at the residual. We therefore sweep the residual from a relative magnitude of 0.02, on the same harness, layers, coherence screen and n as the other site sweeps, with the refusal direction fitted both at the residual itself and at the architecture’s write site.

Table 16 reports it. On Llama-3.1 and the selective-scan SSM the residual defends coherently over a narrow band of relative magnitudes, above which the model degenerates rather than resists, while on Mistral the coherent effect is marginal. RWKV-6 differs. Its block output stays coherent and unmoved across the same band under its own direction and under the time-mix direction alike, and a matched random direction behaves identically. The write site defends over a wider band of relative magnitudes on every architecture we test. Measured at matched *absolute* magnitude, however, the two sites coincide once the direction is held fixed (Appendix A.35), so this band-width gap reflects the norm difference between the sites, not the intervention location.

A.18 A transformer MLP control confirms the write-site mechanism

The write site is the point where an architecture writes a fresh, single-layer contribution into the stream before the residual accumulates it, whether or not that contribution mixes information across positions. The transformer’s position-wise MLP output is a clean test of this: it sits at the same architectural position as the attention output, a fresh contribution written immediately before the residual add, at a comparably small norm, but it processes every position independently. We steer it with the identical protocol used for the attention output and the residual (Appendix A.20): amplify the mean refusal direction at the MLP output by a magnitude relative to its own per-token norm, sweeping the same relative gains, on both transformer anchors.

On Mistral-7B, the MLP output (norm 1.33) reaches 100.0% to 0.0% at $\text{rel}=0.5$, fully coherent, as strong a reduction as the attention output’s own best coherent point (100.0% to 85.0% at $\text{rel}=1$). On Llama-3.1-8B the MLP output again defends (27.5% to 2.5% at $\text{rel}=0.25$, fully co-

Table 16: The residual below the grid floor used elsewhere in this paper. Every residual row reported in the main sweeps is at a relative magnitude of 0.25 or above, which is the smallest magnitude those grids contain. Sweeping beneath it finds attack-success drops of at least 0.2 that stay ≥ 0.9 coherent, in bold. The band closes quickly, and above it the model degenerates rather than resisting, which is why a grid starting at 0.25 reports the site as unmoved. “Direction” is the site the refusal direction was fitted at, which is the push site unless stated otherwise. $n=40$ per cell, single seed. Coherence is the fraction of completions passing the distinct-bigram screen (Appendix A.25).

Model	Direction	Base ASR	rel	ASR	Coherent
Mistral-7B	residual	1.000	0.02	1.000	1.000
Mistral-7B	residual	1.000	0.05	1.000	1.000
Mistral-7B	residual	1.000	0.1	0.925	1.000
Mistral-7B	residual	1.000	0.15	0.675	0.875
Mistral-7B	residual	1.000	0.2	1.000	0.025
Mistral-7B	residual	1.000	0.25	1.000	0.000
Mistral-7B	attention	1.000	0.02	1.000	1.000
Mistral-7B	attention	1.000	0.05	0.975	1.000
Mistral-7B	attention	1.000	0.1	0.725	1.000
Mistral-7B	attention	1.000	0.15	0.600	1.000
Mistral-7B	attention	1.000	0.2	0.600	1.000
Mistral-7B	attention	1.000	0.25	0.875	1.000
Llama-3.1-8B	residual	0.275	0.02	0.125	1.000
Llama-3.1-8B	residual	0.275	0.05	0.075	1.000
Llama-3.1-8B	residual	0.275	0.1	1.000	0.725
Llama-3.1-8B	residual	0.275	0.15	1.000	0.000
Llama-3.1-8B	residual	0.275	0.2	1.000	0.000
Llama-3.1-8B	residual	0.275	0.25	1.000	0.000
Llama-3.1-8B	attention	0.275	0.02	0.225	1.000
Llama-3.1-8B	attention	0.275	0.05	0.200	1.000
Llama-3.1-8B	attention	0.275	0.1	0.425	1.000
Llama-3.1-8B	attention	0.275	0.15	0.150	0.850
Llama-3.1-8B	attention	0.275	0.2	0.125	0.400
Llama-3.1-8B	attention	0.275	0.25	0.350	0.075
Falcon-Mamba-7B	residual	1.000	0.02	1.000	1.000
Falcon-Mamba-7B	residual	1.000	0.05	0.825	1.000
Falcon-Mamba-7B	residual	1.000	0.1	0.900	1.000
Falcon-Mamba-7B	residual	1.000	0.15	0.875	0.850
Falcon-Mamba-7B	residual	1.000	0.2	1.000	0.075
Falcon-Mamba-7B	residual	1.000	0.25	1.000	0.000
Falcon-Mamba-7B	mixer	1.000	0.02	1.000	1.000
Falcon-Mamba-7B	mixer	1.000	0.05	0.975	1.000
Falcon-Mamba-7B	mixer	1.000	0.1	0.925	1.000
Falcon-Mamba-7B	mixer	1.000	0.15	0.700	1.000
Falcon-Mamba-7B	mixer	1.000	0.2	0.775	1.000
Falcon-Mamba-7B	mixer	1.000	0.25	0.875	1.000
RWKV-6	residual	0.475	0.02	0.450	1.000
RWKV-6	residual	0.475	0.05	0.475	0.975
RWKV-6	residual	0.475	0.1	0.500	0.950
RWKV-6	residual	0.475	0.15	0.500	0.950
RWKV-6	residual	0.475	0.2	0.500	0.950
RWKV-6	residual	0.475	0.25	0.500	0.950
RWKV-6	time-mix	0.475	0.02	0.475	1.000
RWKV-6	time-mix	0.475	0.05	0.450	1.000
RWKV-6	time-mix	0.475	0.1	0.450	1.000
RWKV-6	time-mix	0.475	0.15	0.450	1.000
RWKV-6	time-mix	0.475	0.2	0.500	0.975
RWKV-6	time-mix	0.475	0.25	0.500	1.000

Table 17: The MLP output, a non-mixing site, defends as well as or better than the mixing attention output, at a matched relative gain, on both transformer anchors. The residual (Table 15 gives the analogous RWKV numbers, and the transformer residual figures are in Appendix A.20) does not defend anywhere in this range. It defends only below it, over a band too narrow to deploy (Appendix A.17). Coherence is the fraction of completions passing the distinct-bigram screen (Appendix A.25). Each row reports the most attack-success-reducing relative gain at which output stayed at least 90% coherent.

Model	Site	Norm	rel	Base $\rightarrow$ gated ASR	Coherent
Mistral-7B	Attention	0.79	1	100.0% $\rightarrow$ 85.0%	100.0%
Mistral-7B	MLP	1.33	0.5	100.0% $\rightarrow$ 0.0%	100.0%
Llama-3.1-8B	Attention	0.78	0.5	27.5% $\rightarrow$ 0.0%	100.0%
Llama-3.1-8B	MLP	1.69	0.25	27.5% $\rightarrow$ 2.5%	100.0%

herent), with a narrower coherent operating window than Mistral’s: by $\text{rel}=1.0$ the MLP output is already fully degenerate on Llama, where it stayed coherent through $\text{rel}=1.0$ on Mistral. This model-to-model difference in degree matches the variation the attention site and the baseline attack-success rate already show between these two models elsewhere in this paper. What holds on both models without exception is the contrast that defines the write site: the residual does not defend anywhere in this range on either model (Appendix A.20), and where it does defend, below this range, it does so over a band too narrow to deploy (Appendix A.17), while the MLP output, which does not mix across positions, defends cleanly on both.

A.19 The SSM anchor’s own non-mixing site, and a control against mere proximity

The transformer’s MLP control tests non-mixing on a reference architecture. The SSM anchor has its own non-mixing component: the block output $y_t = C_t h_t + D x_t$ (Section 2) splits into a mixed term, $C_t h_t$, routed through the recurrent scan, and a skip term, $D x_t$, that carries no cross-token information at all. We steer only the skip term, with the identical relative-magnitude, open-loop protocol used throughout this appendix, on the same eight layers as the deployed gate ($n=40$ harmful and benign-but-spicy prompts).

At $\text{rel}=2$ the skip term lowers attack success from 100.0% to 17.5%, 98.6% coherent, matching the transformer MLP control’s conclusion on the paper’s primary architecture: a non-mixing site defends. The cost is real, not a calibration artifact: over-refusal climbs steadily with the push, from 13.3% at baseline to 43.3% at the strongest coherent point, and a finer sweep between $\text{rel}=1$ and $\text{rel}=2$ (the 1.25–1.75 rows above) finds no better-calibrated point hiding in between; past $\text{rel}=2$ the site degenerates (17.1% coherent at $\text{rel}=4$).

A second control asks a sharper question: is it specifically the fresh write that matters, or is intervening anywhere near this block sufficient? We steer the mixer’s raw input, before Δ , B , and C are even computed, with the same protocol.

No usable operating point exists. The only fully coherent

Table 18: The SSM’s non-mixing skip term $D x_t$, gain sweep on Falcon-Mamba-7B (prefix injection, base ASR 100.0%, base over-refusal 13.3%). Coherence is the fraction passing the distinct-bigram screen (Appendix A.25).

rel	ASR	Over-refusal	Coherent
0.25	100.0%	20.0%	100.0%
0.5	92.5%	13.3%	100.0%
1.0	77.5%	20.0%	100.0%
1.25	65.0%	33.3%	100.0%
1.5	57.5%	36.7%	100.0%
1.75	32.5%	46.7%	98.6%
2.0	17.5%	43.3%	98.6%
4.0	100.0%	0.0%	17.1%

Table 19: The mixer’s raw input (before selection), gain sweep on Falcon-Mamba-7B (prefix injection, base ASR 100.0%, base over-refusal 13.3%).

rel	ASR	Over-refusal	Coherent
0.25	87.5%	46.7%	100.0%
0.5	57.5%	100.0%	100.0%
1.0	100.0%	0.0%	51.4%
2.0	100.0%	0.0%	87.1%
4.0	100.0%	0.0%	0.0%

setting barely moves attack success ($\text{rel}=0.25$: 87.5%). At $\text{rel}=0.5$, still coherent, attack success falls to 57.5%, but over-refusal reaches 100.0%: every benign prompt tested is refused. Past that point the model degenerates outright (51.4% coherent at $\text{rel}=1$). Proximity to the write-site block is not sufficient; the fresh write itself, specifically, is what matters.

The write site is the point where an architecture writes a fresh, single-layer contribution into the stream. This is an empirical regularity confirmed across every architecture tested, not a claim derived from first principles, and we rule out the three most likely alternative explanations for it rather than assert it directly: that a write site’s effect comes from mixing information across positions, which the non-mixing transformer MLP control (Appendix A.18) and the SSM anchor’s own non-mixing skip term (Appendix A.19) both rule out; that it comes from a smaller activation norm relative to the residual, which the relative-magnitude control rules out (Appendix A.20); or that mere proximity to the write-site block is enough, which the SSM’s mixer-input control rules out (Appendix A.19). A relative-magnitude push to a fresh contribution changes one ingredient the rest of the network combines with others downstream. The same relative push to the residual perturbs the compounded sum of every prior layer’s output at once, which is why the residual degenerates into generic incoherence rather than producing a targeted behavioral shift. RWKV’s time-mix and channel-mix outputs (Appendix A.16), the SSM’s selective-scan block output, the hybrid’s attention and Mamba outputs, and the transformer’s attention and MLP outputs are all fresh sub-block writes by this definition, and every one of them steers. The residual,

the one site that only accumulates, never does.

A.20 The write-read direction defends across architectures

The effective intervention is a *write site*, the point where an architecture writes a fresh, single-layer contribution into the stream before the residual accumulates it, rather than the residual stream itself. The main paper’s Table 6 gives the site and its prefix-injection defense for one exemplar per architecture class. The site is the selective-scan block output in Mamba (the contrast replicates on a second selective-scan SSM, Falcon3-Mamba) and the attention output projection in a transformer. In every case the residual stream is the weaker site.

The pattern is a property of the architecture, not of one model’s training. On Falcon3-Mamba, an independently trained selective-scan SSM, we run the recipe as written (Appendix A.2): gate layers are the top-8 by Δ AUROC, the same set as the gain grid in Figure 6, at its calibrated gain $\alpha=4$. The block-output gate lowers fictional-framing attack success from 56.0% to 4.5% with every gated completion passing the coherence screen and over-refusal held at the undefended 16.0%, matching the grid’s operating point (4.5%). The same detect-then-steer loop at the residual stream of the same layers, at the same gain, leaves attack success at 45.0%, and it stays there at $\alpha=1$ (45.0%, where the under-gained block gate reaches only 32.5%). The residual is therefore partly steerable on Falcon3 but plateaus well above the write site, so the block-defends, residual-lags contrast holds across two separately trained selective-scan models ($n=200$ per cell). Layer selection matters as much as gain here: keeping the anchor’s mixer-output ranking instead picks early layers on Falcon3, and at $\alpha=4$ that set over-steers (only 65.5% of gated completions coherent). Backing that set off to $\alpha=1$ restores coherence (100.0%) and gives a sharper-looking but off-recipe contrast (0.0% versus 94.0%), whereas the Δ -ranked set is coherent at every gain up to 4 (minimum distinct-bigram ratio 0.997). Layer selection and gain are calibrated together, per model, as throughout.

The write-site intervention holds across every architecture class we test. Amplifying the refusal direction at the write site lowers prefix-injection attack success on the Mamba block output (98.0% [95.0, 99.2] to 5.0% [2.7, 9.0], Wilson 95% CIs) and the transformer attention output. On the transformer we use the same eight-layer recipe as the SSM defense, calibrated to a gain that keeps generation coherent: Mistral-7B drops from 98.0% to 14.0% and Llama-3.1-8B from 25.0% to 0.0%, with genuine refusals rather than degenerate text (Appendix A.24). The gain is calibrated per model because a shared default over-steers a steering-sensitive model into incoherence, at which point the attack-success drop is degeneration and not refusal (Appendix A.24). What separates the site from the residual is not defense strength but the site itself: steering the residual stream instead of the write point falls well short of the write site on Falcon3-Mamba above (45.0% against 4.5% at the same layers and gain) and fails outright on the Llama residual, where over-amplifying refusal increases attack success (Li et al. 2026).

Is the mixer-defends, residual-fails contrast a scale artifact? The residual accumulates every block’s output, so its activations have a larger norm than the mixer’s at the same layer (13.8 vs 1.8, a $7.8\times$ ratio). A fixed absolute gain is therefore a smaller relative perturbation at the residual. We rule this out with the same norm-matched control used for RWKV-6 (Appendix A.16), steering each site at a magnitude relative to its own per-token activation norm. At a relative push of $1\times$ the mixer’s own norm, attack success falls from 100.0% to 0.0% with output 97.5% coherent, a genuine defense. The residual gets no chance to show whether a larger push would defend it: at the *smallest* relative magnitude we test ($0.25\times$ its own norm, already a larger absolute push than the mixer’s operating point), output already degenerates into incoherent text (0.0% coherent) while attack success stays at 100.0%. This is the same failure pattern the mixer itself shows once over-driven, and the same pattern seen at the residual in RWKV. At this relative magnitude the residual has no coherent operating window in which to defend, while the mixer does. The matched-absolute control (Appendix A.35) shows this reflects the norm gap between the sites, not an intrinsic property of the location.

The same control on the transformer (Mistral-7B, attention output versus the whole decoder-layer output, the same closed-loop gating as the established alpha-2 defense above) gives the same qualitative result with a narrower margin. The residual’s activations have norm 22.0 against the attention output’s 0.79, a $27.9\times$ ratio, and at the smallest relative push we test the residual already degenerates into incoherent text with attack success unchanged at 100.0%, the same pattern as the SSM and RWKV residuals. The attention output does defend, from 100.0% to 85.0% at a relative push of $1\times$ its own norm while 100.0% coherent, but the drop is far shallower than the mixer’s, and a larger relative push degrades it back toward the baseline rate before output stops being coherent. The write-versus-residual contrast holds in every architecture we test, though how strong a defense the write site itself affords at matched relative magnitude does not.

A.21 The read-site pattern holds in a hybrid model

The shared-representation result (Section 4) extends to Zamba2-7B as well: a transformer-read refusal direction transports into its residual stream by the same orthogonal map at AUROC 1.00, matching the target’s own probe (1.00) and above its shuffled control 0.52, so the refusal representation is present and shared in the hybrid too, not only in the pure SSM and RWKV-6.

Zamba2-7B-Instruct is a Mamba2 backbone with periodic shared-attention layers, so a block has both a Mamba mixer write point and, at the hybrid layers, an attention write point, on top of the residual stream. This is the case the “re-point, not rebuild” claim leans on, where it is not obvious a priori which write site to steer. On prefix injection the base model is jailbroken at 31.7% [21.3, 44.2], a meaningful attack. Amplifying the refusal direction read at the attention write site defends, lowering attack success to 0.0% [0.0, 6.0], while a direction read from the residual stream does not (36.7%).

The Mamba mixer write site also defends, but only under calibration: sweeping its layers and gain, an early-layer band at gain 16 lowers attack success from 27.5% [16.1, 42.8] to 5.0% [1.4, 16.5] with coherent refusals (Wilson 95% CIs), while a mid-layer band over-steers and raises it to 40.0%. Both write sites therefore defend and the residual does not, so the same read-site pattern holds in a hybrid, and which write site is easiest to use is a per-model calibration choice, here the attention output.

A second, smaller checkpoint replicates the site, not the margin. Zamba2-2.7B-Instruct-v2 (54 layers vs. 81, a distinct pretraining run in the same family, not a second independent lab) was tested at the same five-layer band, steered at a magnitude relative to each site’s own per-token norm rather than the 7B model’s fixed gain, since the two are not directly comparable in absolute terms (Appendix A.20). At $n=60$, the Mamba mixer write site lowers attack success from 98.3% to 81.7% at $\text{rel}=1.0$, coherent (96.7%), a smaller margin than the 7B model’s. The residual’s own lower reading at the same relative magnitude (98.3% to 93.3%) is not a genuine defense, output there is only 0.0% coherent, the same degeneration-not-defense pattern the residual shows everywhere else in this paper. The attention write site found no relative magnitude in the range we swept that both defended and stayed coherent. The write-versus-residual contrast holds at this smaller scale, though its strength does not.

The 7B residual, tested with its own direction, fails the same relative-magnitude sweep. The fixed-alpha result above uses one direction fit at the attention site and pushed everywhere, which leaves open whether the residual simply lacks a good direction of its own. We fit the refusal direction separately at the residual, matching the norm-scale-controlled protocol used for the SSM, the generic transformer, and Llama, and swept it at $n=60$. At the smallest push ($\text{rel}=0.2$) attack success briefly drops from 36.7% to 1.7%, coherent (100.0%), but the effect reverses almost immediately: by $\text{rel}=0.3$ attack success is back up to 88.3%, and the completions are still 98.3% coherent, a genuine capitulation to the attack rather than broken text. Unlike either write site, the residual has no stable operating point across the range swept.

A.22 Causal ablation of the transported direction: attention is modest, the Mamba mixer dominates

The rotation-transport test above establishes that the refusal representation is shared with the hybrid. Whether *removing* the transported direction causally jailbreaks Zamba2, the test the shared- representation claim rests on for the pure SSM (Section 4, App. A.9) and RWKV-6 (App. A.16), was checked at both write sites, with an unequal result.

Attention site: a real, modest, fully coherent effect. Fitting a standardized Procrustes rotation between Llama’s residual and Zamba2’s attention write site (the two tied attention-block instances, disambiguated by call order since they alias across all 13 hybrid layers), transporting Llama’s refusal direction through it, and suppressing that direction

at the same gain established for amplifying it there (gain 8, Appendix above) on held-out plain harmful prompts ($n=60$ per seed, pooled over 3 seeds, $n=180$) raises attack success from 8.9% to 16.7% [11.9, 22.8], against a matched orthogonalized-random control at 8.9% [5.5, 14.0]. Even pooled, these Wilson intervals still just overlap, so the attention-site effect, while directionally consistent with the Mamba-site result below, is not on its own statistically decisive. We do not lean on it in isolation, only in combination with the much larger and cleanly-separated Mamba-site effect. This test asks whether a *foreign, transported* direction is specifically causal, a different and stronger claim than Table 6’s defense number, which amplifies the model’s own refusal direction rather than a transported one and is unaffected by this test’s more modest margin. Scored with Llama-Guard-3-8B throughout (unlike RWKV above, this barely differs from the lexical heuristic). Output stays fully coherent (distinct-bigram and repetition screen, Appendix A.25) throughout.

Mamba site: a confound at the established gain, resolved by a coherence-gated sweep. The same transport-and-suppress test at the Mamba mixer site (the same early-layer band as above, independent per-layer modules, no tying issue), first run at the gain already established for *amplifying* refusal there (gain 16, Appendix above), gave attack success 100.0% for the transported direction and 85.0% for the matched random control, at output coherence 68.3% and 41.7% respectively. The random control degrading *more* than the transported direction, both well below full coherence, is the signature of an over-strong intervention breaking generation regardless of direction, not a specific causal effect: a gain calibrated for amplifying refusal is not automatically the right gain for suppressing it.

We therefore swept the suppression gain and tracked coherence for both the transported and the matched random direction at each point, taking the largest gain at which the transported direction’s output stayed at or above 80.0% coherence as the operating point (mirroring the coherent-gain search used for RWKV’s write sites, App. A.16). Table 20 gives the full sweep, with a Wilson 95% CI on every cell ($n=60$ per cell throughout). At gain 8, both cells are still fully coherent (100%) and already show a large, non-overlapping gap (60.0% [47.4, 71.4] transported against 5.0% [1.7, 13.7] random). This is the threshold- robustness check for our choice of 80.0% as the coherence cutoff: the qualitative conclusion, a large and specific transported-versus-random gap, already holds at gain 8 under the strictest possible coherence requirement (100%, i.e. every completion passes), so it is not an artifact of setting the cutoff at exactly 80.0% rather than some other value. At the chosen operating point (gain 12, transported coherence 88.3%), attack success reaches 91.7% [81.9, 96.4] for the transported direction against 25.0% [15.8, 37.2] for the random control, which remains fully coherent (100.0%). The two CIs do not overlap. Past this point (gain 16 and above) both cells degenerate together and the gap becomes uninterpretable, exactly the confound the initial gain-16 run exhibited.

This holds across seeds. Repeating the gain-12 check with

two further seeds (each drawing its own prompt split and random-control direction) reproduces a large, coherent gap in the same direction both times, with the random control never approaching the transported cell. Pooled over all three seeds ($n=180$, Llama-Guard-3-8B): transported 93.3% [88.7, 96.1], random 22.8% [17.3, 29.4], non-overlapping.

We find that the Mamba mixer, not the attention block, is therefore where Zamba2’s refusal control actually concentrates: the clean causal-ablation effect there (93.3% pooled) is larger than RWKV-6’s (54.7% against a random control at 32.3%, a narrower margin, Appendix A.16) and substantially larger than the attention site’s own effect (16.7% against 8.9%). The earlier, attention-only reading of a "weak" hybrid effect reflected an incomplete write-site search, not a property of the hybrid’s refusal mechanism.

A candidate mechanism: Zamba2 is architecturally Mamba-dominated. Why does the same hybrid favor attention for amplification but Mamba for ablation? We measured the native harm-minus-benign direction’s norm at the attention site and at 68 sampled Mamba layers, each relative to that site’s own mean activation norm (scale-invariant, so sites at different depths and with different activation scales are comparable). Per instance, Mamba’s mean relative signal (0.37) is comparable to attention’s (0.28). The difference is where the two sites fire. Attention exists at only 13 of Zamba2’s 81 layers, both tied to just 2 shared weight instances; Mamba mixers exist at all 68, each independently parameterized. Weighting the per-instance signal by how many times each component type actually fires gives a cumulative estimate of 3.6 for attention against 24.9 for Mamba, a $6.9\times$ difference. Zamba2 is, by layer count, mostly a Mamba model with attention sprinkled in sparsely, and the harm-relevant signal appears to accumulate accordingly: most of it lives in Mamba simply because most of the network’s depth is Mamba, so removing it there removes more of the model’s total refusal-relevant computation than removing it from the sparse, tied attention component. This is consistent with, but does not by itself explain, why attention is still the easier site to calibrate for amplification: its two tied instances behave identically wherever they fire, while the sampled Mamba layers vary five-fold in relative signal across depth (0.37 mean, individual layers ranging from 0.11 near the input to 0.57 mid-network), which is a more natural explanation for why a single gain calibrated for one Mamba layer over- or understeers at another (Appendix above), and why attention’s homogeneity makes it the easier defense lever. We report this as a candidate mechanism, not a proven one: it is consistent with both observations but was not tested by directly intervening on the signal magnitude itself.

A.23 The persona fix does not generalize: RWKV and Zamba2 each fail a different way

The two-direction persona fix (Appendix A.14) is calibrated on Falcon-Mamba: amplifying refusal alone barely moves attack success, ablating the persona direction alone backfires, and only the combined push defends. We test the same two-direction fix, at each architecture’s own established write site, on RWKV-6 (time-mix output, gain 16) and Zamba2 (Mamba

mixer, gain 16). Neither replicates the Falcon-Mamba pattern, and the two failures are different in kind.

RWKV: the wrapper evades the detector, not just the fix. Applied closed-loop (gated on the detector, as deployed), all four conditions, base, refusal-only, persona-only, and combined, are byte-identical: the gate never fires on the DAN-wrapped evaluation set. The same harmful prompts, plain, fire the detector at 100.0%; DAN-wrapped, they fire at 0.0% (mean probe: we transport a detector fit on Llama-3.1-8B’s residual (AUROC 1.00 on its own harm-versus-benign split) into RWKV’s time-mix space by the same standardized-Procrustes rotation behind the shared-representation claim, and it is evaded just as completely (plain 0.994, wrapped 0.000). A detector trained on a different architecture entirely, with no exposure to RWKV’s activations beyond the rotation fit, misses the same wrapped prompts the native probe misses, so the wrapper is evading detection at the level of what it does to the representation, not exploiting one classifier’s decision boundary.

Bypassing the gate to apply each condition unconditionally isolates whether the fix itself would work if triggered. It would not, and the pattern is the reverse of Falcon-Mamba’s: refusal-amplification alone helps (19.0% to 8.0%), persona-ablation alone is the term that backfires, sharply (to 77.5%), and the combined push lands at 57.0%, above the unablated baseline and with the gate’s benign over-refusal cost also rising (marginal 8.0%). On Falcon-Mamba persona-ablation is the fix and refusal-amplification is harmless. On RWKV it is the opposite term that backfires and the one that helps is not enough on its own.

Zamba2: the gate fires, but the fix does not help. At the Mamba write site the detector is not the problem: refusal and combined gated completions differ from the ungated baseline on 60 of 60 harmful prompts. At the site’s calibrated operating point ($\alpha=16$, the same gain used for the headline Mamba-site defense), refusal-amplification alone still backfires, though mildly (0.0% to 5.0%). This is gain-sensitive rather than a general over-amplification failure: an earlier, uncalibrated pass at $\alpha=12$ backfired far worse (to 26.7%), so that larger gap was a calibration artifact of the wrong gain, not evidence this write site over-amplifies the way a transformer’s residual does. Persona-ablation alone is inert at this operating point too: every one of its 60 gated completions is byte-identical to the ungated baseline (0 differ). The combined fix reaches 8.3% (98.3% coherent), better than refusal alone but still above doing nothing at all.

"Inert" describes persona-ablation tested in isolation, not its role once paired with a strong enough refusal term. A grid over refusal gain $\alpha \in \{2, 4, 6, 8, 12\}$ and persona-ablation gain $\beta \in \{0, 1, 2\}$ ($n = 60$ judged completions per cell) shows every cell at $\alpha \leq 6$ flat at the base rate regardless of β . The fix only engages once refusal-amplification is already strong enough to backfire on its own. At $\alpha = 12$, raising β from 0 to 2 pulls attack success from 26.7% down to 18.3% (98.3% coherent), a real, coherent partial mitigation, not the null effect the isolated test suggests. It does not reach the unablated baseline at any grid point tested.

Table 20: Coherence-gated suppression-gain sweep at Zamba2’s Mamba mixer write site. Both the rotation- transported refusal direction and a matched orthogonalized-random control are suppressed at each gain; coherence is the fraction of outputs passing the distinct-bigram/repetition screen. The chosen operating point (bold) is the largest gain at which the transported direction stays at or above 80.0% coherence. Base attack success (gain 0) is 11.7%. Gains 8, 12 are rescored with Llama-Guard-3-8B (matching the operating-point number quoted in the main text). The remaining gains are the original lexical-heuristic score, not rejudged, and are shown only to establish the coherence frontier’s shape, not as headline attack-success numbers.

Gain	Transported ASR	Transported coh.	Random ASR	Random coh.
1	13.3%	100.0%	13.3%	100.0%
2	11.7%	100.0%	15.0%	100.0%
4	20.0%	100.0%	13.3%	100.0%
8	60.0%	100.0%	5.0%	100.0%
12	91.7%	88.3%	25.0%	100.0%
16	100.0%	68.3%	85.0%	41.7%
24	100.0%	26.7%	98.3%	66.7%

Summary. The two-direction persona fix, needed once refusal is steerable with a single direction to begin with, is a separate attack-specific augmentation, not one of the five recipe steps in Section 2.5. It is not simply untested beyond Falcon-Mamba and Mistral. It is tested, and it does not transfer: RWKV fails upstream of the fix, at detection, while Zamba2’s detector works but the fix’s two terms swap roles from Falcon-Mamba and still net worse than an unmodified model. A fix validated on one architecture’s own geometry should not be assumed to carry over even when the qualitative readout principle it augments does.

A.24 A coherent operating point on the transformer attention site

The transformer write-site intervention in Table 6 requires a calibrated gain. Too small a gain leaves attack success unchanged, and too large a gain pushes generation into degenerate text, so an uncalibrated attack-success drop can be degeneration rather than refusal. We sweep the gain α and screen every completion for degeneration: a completion is coherent if its distinct-bigram ratio is at least 0.5 and no single word is more than 30% of its tokens. Table 22 reports Llama-Guard-3 attack success and the coherent fraction on prefix injection ($n=100$ harmful, 60 benign-but-spicy).

A coherent operating point exists on both transformers at $\alpha=2$. Llama-3.1-8B falls from 25.0% [17.5, 34.3] to 0.0% [0.0, 3.7], and Mistral-7B from 98.0% [93.0, 99.4] to 14.0% [8.5, 22.1], both fully coherent on harmful and benign prompts (Wilson 95% confidence intervals). Degeneration appears only at $\alpha \geq 3$. Mistral, whose base attack success is much higher, degrades one step earlier than Llama, which is why its gain is calibrated separately. We also tested Qwen2.5-7B, but our attacks do not jailbreak it (its base attack success is near zero), so it does not serve as a test of the defense and is excluded from the main comparison.

The closed-loop trigger is what makes this safe to deploy. Because the detector fires on every harmful prompt and no benign one, the closed-loop gate and the open-loop DeTAM-style edit (Li, Jiang, and Wei 2025) reach the same attack success, so they share the gated-ASR column. They diverge on the benign side. At the coherent operating point $\alpha=2$ the closed-loop gate holds benign over-refusal at the base rate

(10.0% on Llama, 11.0% on Mistral), while the open-loop edit steers every prompt and raises it to 87.0% and 69.0%. Pushed to $\alpha=4$ the open-loop edit additionally collapses benign coherence (to 53.3% on Llama and 20.0% on Mistral), while the closed-loop gate leaves benign prompts untouched. This over-refusal advantage holds across all three attack families (Table 21): on the coherent $\alpha=2$ gate the closed-loop and open-loop edits reach the same attack success on each attack, but the closed-loop gate keeps benign over-refusal at the base rate while the open-loop edit raises it to 87.0% on Llama and 69.0% on Mistral.

Table 21: DeTAM comparison at the coherent gain ($\alpha=2$, $n=200$ harmful, 100 benign per attack, an independent draw from the gain-sweep evaluation set in Table 22, so base rates differ slightly by sampling variance between the two tables). The closed-loop gate and the open-loop DeTAM-style edit reach the same gated attack success on each attack (the detector fires on all harmful prompts), so the difference is over-refusal: the closed-loop trigger holds it at the base rate, the open-loop edit raises it sharply. On Mistral, single-direction amplification is not reliable against persona attacks: role-play attack success rises from 63.5% to 78.0% (it backfires), and fiction falls only partially, from 96.5% to 81.5%. This is the same attack-dependence as the SSM, which the two-lever gate that also suppresses the persona defends (Appendices A.14 and A.15).

Attack	Base ASR	Gated ASR	Over-ref. (ours)	Over-ref. (DeTAM)
<i>Llama-3.1-8B</i>				
Prefix injection	28.0%	2.0%	10.0%	87.0%
Persona roleplay	0.5%	0.0%	10.0%	87.0%
Fictional framing	19.5%	1.0%	10.0%	87.0%
<i>Mistral-7B</i>				
Prefix injection	96.0%	15.0%	11.0%	69.0%
Persona roleplay	63.5%	78.0%	11.0%	69.0%
Fictional framing	96.5%	81.5%	11.0%	69.0%

The attention-site defense itself is strongest against prefix injection, where it lowers attack success on both transformers (Table 6), and weaker on persona roleplay and fictional framing, the same attack-dependence we report at the SSM

block output (Appendix A.14). On Mistral the roleplay case does not improve over the base rate, so we scope the transformer defense claim to prefix injection, where the read-site pattern is clean on every architecture we test.

Table 22: Attention-site gain sweep on two transformers (prefix injection). Gated ASR (Llama-Guard-3) is shared by the closed-loop gate and the open-loop DeTAM-style edit, since the detector fires on all harmful prompts. “Harm coh.” and “Benign coh.” are the coherent fractions under our gate. “DeTAM benign” is the open-loop edit’s benign coherence, which collapses at high gain.

α	Gated ASR	Harm coh.	Benign coh.	DeTAM benign
<i>Llama-3.1-8B</i> , base ASR 26.0%				
0.5	7.0%	100.0%	100.0%	100.0%
1.0	2.0%	100.0%	100.0%	100.0%
1.5	1.0%	100.0%	100.0%	100.0%
2.0	1.0%	100.0%	100.0%	100.0%
3.0	3.0%	80.0%	100.0%	100.0%
4.0	0.0%	63.0%	100.0%	53.3%
<i>Mistral-7B</i> , base ASR 97.0%				
0.5	98.0%	100.0%	100.0%	100.0%
1.0	88.0%	100.0%	100.0%	100.0%
1.5	39.0%	100.0%	100.0%	100.0%
2.0	15.0%	100.0%	100.0%	100.0%
3.0	78.0%	89.0%	100.0%	98.3%
4.0	87.0%	40.0%	100.0%	20.0%

A white-box gradient attack. The jailbreaks above are templated. To test the transformer attention-site gate against an adversary with white-box access, we run Greedy Coordinate Gradient (Zou et al. 2023b), fitting a per-prompt adversarial suffix on each of 100 held-out harmful behaviors by back-propagating an affirmative-target loss to the one-hot suffix tokens (Mistral-7B, batched candidate search, no external dependency). We fit a length-20 suffix per behavior over 500 optimization steps, selecting from the top 256 gradient-ranked substitutions with 256 candidates per step and the affirmative target “Sure, here is”. The fitted suffixes reach base attack success 41.0% (Llama-Guard-3, Wilson 95% confidence interval [31.9, 50.8]) on Mistral-7B-Instruct-v0.2, a competitive white-box attack. The detector fires on all 100 attacked prompts, and the closed-loop attention-site gate at the coherent gain lowers attack success to 18.0% (Wilson 95% confidence interval [11.7, 26.7]) with genuine refusals and benign over-refusal unchanged at 11.0%, so the gate more than halves a competitive gradient-optimized attack rather than being evaded by it, with the two intervals disjoint. Because GCG optimizes its suffix toward the affirmative target “Sure, here is”, a judge that keyed on the opening rather than the content could score a steered refusal as compliance. It does not: the out-of-family Gemma judge, which reads the whole response, independently reports the same reduction (16.0% to 5.0%), and the gain used here is the coherent one, since a higher gain degenerates the text and the forced affirmative opening then survives into the judge’s view. This reproduces across seeds, not only the one reported

above: pooled over three seeds ($n=300$), the gate lowers attack success from 43.3% to 18.7% (17.7% to 3.7% under the out-of-family judge). We report GCG on Mistral because it is the transformer where the attack lands: at the same 500-step budget the same configuration barely jailbreaks the more heavily safety-tuned Llama-3.1-8B (base attack success 0.0%) and Qwen2.5-7B (base attack success 6.7%), so only Mistral gives a gradient attack strong enough to exercise the gate. We run GCG on a transformer rather than the SSM because the fused selective-scan kernels used at inference are non-differentiable. This is a speed choice, not an absence of a differentiable path: Falcon-Mamba’s own fallback forward (the one every experiment here actually runs, since the fused kernels are unavailable in our environment) is plain PyTorch with no detached operations, and `mamba_ssm` ships a differentiable reference scan (`selective_scan_ref`) for exactly this purpose. We did not adapt GCG’s candidate-search loop to run through this much slower path. The gradient-free optimized suffix in Section 5.3 is the SSM counterpart of this attack.

A gate-aware adaptive adversary. The suffix above is optimized against the plain model. The stronger threat model optimizes it against the intervention. We install the write-site amplification as an always-on gate before the search begins, so every forward pass, both the affirmative-target gradient and the candidate re-ranking, runs through the amplified model. The amplification $h \mapsto h + \alpha u$ is differentiable, so the gradient flows into the suffix and the attack knows the refusal direction, the layers, and the gain. A run-time check confirms the installed gate changes the affirmative-target loss before the search starts, so the optimization cannot silently reduce to the plain attack. We fit 100 suffixes at the same length, budget, and coherent gain as above, and evaluate three conditions with the same four-judge panel, the ungated model, the deployed selective gate, and the always-on gate in isolation.

Optimizing against the gate does not make the suffix more harmful. Its ungated attack success is 42.0% on Llama-Guard and 16.0% on the out-of-family whole-response judge, matching the non-adaptive suffix (41.0% and 16.0%), so the adversary gains nothing on the plain model. The always-on intervention is not defeated either. It holds the out-of-family attack success at 5.0% (Wilson 95% confidence interval [2.2, 11.2]), statistically indistinguishable from the non-adaptive deployed gate’s 5.0% [2.2, 11.2] (identical point estimate and Wilson interval at matched n), so a suffix built against the mechanism does not beat the mechanism. The always-on gate over-refuses benign prompts at 69.0%, which is why the deployment is selective and the always-on variant is a diagnostic rather than a defense.

The adaptive suffix’s only success is against the detector *trigger*. On the deployed selective gate the primary-judge reduction the non-adaptive attack produced (41.0% to 18.0%) disappears, to 44.0% ([34.7, 53.8]). That number is the affirmative-opening artifact this appendix already flags on the primary judge, since the out-of-family judge that reads the whole response still shows the deployed gate lowering attack success from 16.0% to 12.0% ([7.0, 19.8]). The

honest erosion is on that trusted judge, where the deployed rate rises from the non-adaptive 5.0% to 12.0%, because the adaptive suffix lowers the per-layer detector scores and the selective gate then amplifies at fewer layers. Pooled over three seeds ($n=300$), the same pattern holds: the deployed selective gate’s out-of-family rate rises from the non-adaptive 3.7% to 15.3% under the adaptive suffix, while the always-on gate in isolation still holds attack success at 2.3%. This complements the probe-evasion attack of Appendix A.28, which optimizes directly against the detector, and reaches the same conclusion from the opposite direction: the attackable surface is the detector that triggers the gate, not the write site that carries the defense.

A.25 Coherence and gain-calibration checks

Two validity checks run under every defense number we report.

Coherence. Activation steering can push generation off distribution into degenerate text, a single repeated token or a collapse in vocabulary, and our attack-success metric counts any non-refusing response as a success, so degenerate output would inflate it. We therefore coherence-screen gated completions during gain calibration: a response whose most frequent token exceeds 40% of its tokens, or whose distinct-token ratio falls below 0.3, is treated as degenerate rather than as a successful attack. The reported defenses use gains chosen to pass this screen, and we spot-checked the headline gates directly (Falcon-Mamba to 5.0% and the Falcon3 control), where they produce genuine refusal text rather than degeneration.

The RWKV and Zamba2 causal-ablation work (App. A.16, A.22) and the transformer attention sweep (App. A.24) instead use a second, independently-developed screen: distinct-bigram ratio at least 0.5 and no single word more than 30% of tokens. The two definitions were never cross-checked against each other before this pass. We ran both over every raw completion saved from this session’s causal-ablation experiments ($n=750$) and they agree 88.1% of the time. The disagreements are concentrated in phrase-level repetition (e.g. “the product is that the product is” repeated), which the bigram screen catches and the token screen, checking only single-token frequency, does not, and in very short completions, where the bigram screen’s four-word minimum treats brevity itself as a failure the token screen does not. Neither definition dominates the other. We do not retroactively rescoring the main defense numbers under the bigram screen, since 88.1% agreement means the two screens could in principle disagree about a small fraction of individual gated completions, though we have no evidence this changes any reported aggregate rate by more than a percentage point. We did check the one place a screen choice could change a conclusion rather than a percentage point: the Zamba2 Mamba-site operating-point selection (Appendix A.22), which picks the largest gain at which output stays coherent. Recomputing that selection under the token screen instead of the bigram screen, across all three seeds, picks the same gain (12) either way.

Per-model gain calibration. The steering gain is calibrated for each model, because a single shared gain is unsafe.

Table 23: The anchor’s gate at 256 new tokens (attack success / fraction coherent, $n=200$ per cell, Llama-Guard judge). The gain calibrated at 64 tokens ($\alpha=4$) neither stays coherent nor fully defends at 256, and no swept gain does both.

Gain α	Prefix injection	Persona roleplay
base (no gate)	98.0% / 99.5%	82.5% / 100.0%
1	99.0% / 99.0%	89.5% / 99.5%
2	98.0% / 99.5%	91.5% / 98.5%
4 (deployed at 64)	17.5% / 10.5%	55.0% / 23.5%
6	48.5% / 0.0%	60.5% / 0.0%

The gain that defends Falcon-Mamba over-steers the more sensitive Falcon3-Mamba into backfire-compliance and its residual into degeneration, and over-amplifying refusal in an already-robust transformer increases attack success (Li et al. 2026). We calibrate with a single-layer gain sweep read against the coherence screen, taking the gain that minimizes coherent attack success before the degeneration cliff. Reported defenses use the calibrated gain, and the controls above are run at it.

A.26 Generation-length sensitivity of the deployed gate

The defense numbers in this paper are measured at 64 new tokens. Regenerating the anchor’s headline cells at 256 new tokens (Table 23, $n=200$ per cell) leaves the base rates and the benign side unchanged, with over-refusal at the undefended rate in every cell, but raises the gated rates to 17.5% under prefix injection and 55.0% under persona roleplay: past the 64-token horizon the disclaimer-then-comply drift of Appendix A.14 resumes, and the deployed gain’s completions also fail the coherence screen (10.5% and 23.5% coherent). Re-sweeping the gain at 256 tokens finds no coherent defending operating point. Gains at or below 2 are fully coherent but leave attack success at the base rate, and gains at or above 4 lower attack success only at broken coherence, outside the operating regime the coherence screen accepts (Appendix A.25); at gain 6 the output is fully degenerate and judge agreement itself falls to 0.77. The always-amplified steering arm is therefore length-bounded. The detector is not: it fires once, on the prompt, independent of generation length, so at long generations the deployment option the frontier comparison already prices, detect-then-refuse with the same detector, which the gate ties (Appendix A.12), is unaffected.

A.27 A harm-weighted pool closes the dilution attack

The dilution attack appends benign sentences to drag the mean-pooled harm score below the detector threshold. The naive threshold fires through two appended sentences (attack success 2.5%), degrades at four (0.81 firing, 19.0%), and is defeated at six to eight (firing 0.30 and 0.06, attack success 67.0% and 90.0%). The failure is the threshold and not the features: the harm logit falls from 7.86 to -3.8 while the benign cluster stays far below at -23.69, so the two classes stay

separable at every level. Two facts close it. First, recalibration alone is more robust than it appears: the mean harm score does not fall without bound as benign text is appended, it asymptotes to the mean score of the benign filler, which sits above the benign-calibrated threshold. At a 1.0% benign false-positive rate the recalibrated mean-pool detector fires on every prompt we test through 256 appended sentences, its score plateauing near 0 against a threshold of -18. A crafted filler of strongly anti-harmful text could in principle push the mean below threshold, which is the residual concern. Second, that concern is removed by a harm-weighted pool, $s_{\text{pool}} = \sum_t \text{softmax}(\tau s)_t s_t$, which weights each token by its harm logit s_t rather than by position. Low-scoring padding receives near-zero weight, so the score is invariant to padding amount and content: it holds at 57 through 256 sentences (threshold 6), where the mean has long since drifted, at the same false-positive rate. The harm-weighted pool is therefore a drop-in detector that is dilution-invariant by construction. Because harm is cleanly separable on undiluted prompts, switching to it leaves the headline defense and over-refusal unchanged. Only the dilution robustness improves.

A.28 Probe-evasion, and why the harm-weighted trigger closes it

A white-box adversary can optimize a suffix directly against the frozen detector rather than against the model. We fit a gradient-free suffix that minimizes the mean-pooled detector’s prompt-level harm score on held-in harmful prompts, then transfer it to the eval set. The search is a greedy coordinate descent over suffix token positions with batched candidate scoring, keeping the best candidate per round (150 rounds of 64 candidates), so it does not depend on autograd through the fused selective-scan kernels. Against the deployed mean-pooled probe it succeeds: the detector fires on every clean harmful prompt (1.00) but none once the suffix is appended (0.00), so the gate never triggers and attack success rises from 6.7% to 95.0%, the undefended rate.

The break is a property of mean pooling, not of the write site. The suffix works by appending anti-harm tokens that drag the mean harm score down, from a median of 7.8 on clean harmful prompts to -7.7 with the suffix. The harm-weighted pool of Appendix A.27 weights each token by its own harm logit, so those appended tokens receive near-zero weight while the causally-earlier harmful tokens keep their scores. Its score is essentially unchanged by the suffix, holding at a median of 45.7 against 46.0. Calibrated to the same 1% benign false-positive rate, the harm-weighted pool fires on 1.00 of the evasion prompts, as does the max pool (1.00). The harm-weighted and max pools coincide here because the softmax concentrates on the peak harmful token, and the harm-weighted pool’s advantage over a hard max is the graded dilution robustness of Appendix A.27, not this evasion test. An appended suffix cannot lower a pool that weights the fixed harmful span, so the suffix-evasion route is closed by the trigger we already deploy against dilution. Even the mean pool catches the suffix once its threshold is benign-calibrated rather than set at the sigmoid midpoint: the evasion drives the score only to -7.7, still above the benign cluster, so it fires on 1.00 of evasion prompts at the same false-positive rate. We

also fit a suffix directly against the harm-weighted pool. After 200 rounds it still cannot evade it: the pool fires on 1.00 of the attacked prompts at the same false-positive rate, and its score does not fall (median 45.7 to 46.0), because an appended suffix cannot lower the fixed harmful span’s contribution. Only an adversary who rewrites the harmful content itself, and so changes the request, is outside this test. A concurrent activation probe for Gemini max-pools per-token scores so a short harmful span is not diluted by long benign context (Kramár et al. 2026). Our harm-weighted pool goes further on the axis that matters here: it is dilution-invariant by construction against a named adaptive attack, weights tokens by a continuous softmax over harm logits rather than a hard max, and operates inside a closed-loop steering defense rather than as a standalone detector.

A.29 Detector hardening against wrapper text

A detector trained on plain harm can fire on the jailbreak wrapper rather than the harmful request. A benign-but-spicy prompt wrapped in the same jailbreak template trips the plain detector 0.78 of the time, which would over-refuse benign requests that merely carry jailbreak boilerplate. Training the detector with wrapped-benign prompts as negatives removes the confound: the wrapped-benign firing rate falls to 0.00 while harm detection is preserved (wrapped-harm 1.00, plain harm 1.00). The SSM and transformer defenses reported here use this hardened detector. The hybrid runs use a plain harm-versus-spicy detector, since at one of its two sites the hardening does not work (Appendix A.37). Hardening does not change attack success, because both detector variants fire on every wrapped harmful prompt, so it affects only what a gate does to wrapped benign traffic.

A.30 The intervention must use the model’s refusal geometry

The same readable harm direction has opposite effects depending on how it is used (Table 24). Amplifying it toward refusal at the block output lowers attack success from 98.0% to 5.0%, whereas projecting it out of the residual stream barely moves attack success (95.0%). Conversely, on plain harmful prompts that the instruct model mostly refuses (base attack success 10.5%), removing the direction raises attack success to 41.5%: the harm direction is the refusal trigger, so deleting it ablates refusal while amplifying it induces refusal. Steering along isolated subspace components away from the mean refusal direction instead does not control refusal and degenerates the output at the magnitudes needed to defend, so the standard mean refusal direction is what controls refusal at the write site.

A.31 Detector-class ablation: the site contrast is not linear-probe-specific

The write-site result holds when the linear trigger is replaced by a nonlinear one. Swapping the detector for a one-hidden-layer MLP, with the intervention site and the mean-direction gate held fixed, leaves the prefix-injection site contrast unchanged: the block-output gate lowers attack success to 6.7% (from 98.3%), while CAST at the residual stays at 100.0%,

Table 24: The action matters: the same refusal direction (all on Falcon-Mamba-7B-Instruct, $n=200$). Amplifying it at the block output defends, projecting it out of the residual does little, and removing it from a model that was refusing jailbreaks the model. The first two rows use prefix-injected prompts (high base attack success, room to fall). The third uses plain harmful prompts (low base, room to rise).

Action on the refusal direction	Base	After
Amplify at block output (ours), under attack	98.0%	5.0%
Project out of residual, under attack	98.0%	95.0%
Remove direction, plain harm	10.5%	41.5%

matching the linear-detector cells. This contrast is a property of the direction, not of the probe class.

The residual gap also survives CAST’s own gain sweep. Sweeping CAST’s gain over $\alpha \in \{2, 4, 8, 16\}$ under prefix injection gives attack success 41.0%, 100.0%, 97.5%, and 98.0%. Its self-tuned best operating point ($\alpha=2$, chosen by lowest attack success) reaches 41.0% at 12.0% over-refusal, still far above the write-site gate under the same detector, so the residual failure in Table 9 is not an artifact of evaluating CAST at one gain.

A.32 Robustness of the defense

The main tables report 200 prompts per condition. The comparison holds across seeds, under a larger sample, judges from other model families, and a different harmful-behavior distribution. Refitting the refusal direction and redrawing the eval set across 3 seeds leaves the defense stable, with pooled gated prefix-injection attack success 3.3% ([2.2, 5.1], $n=600$) against a base of 98.2% ([96.7, 99.0]), non-overlapping. Replicating the headline and the site-and-loop cells at 300 harmful prompts per condition reproduces the defense under prefix injection (attack success 96.3% to 4.0%, Wilson 95% confidence interval [2.3, 6.9]). Beyond the two Llama-family judges, which agree on 93.5% of cases, two judges from different families confirm the gate rather than leaving a within-family artifact, a prompted Mistral-7B-Instruct-v0.2 (gated 1.5% against a base of 96.0%) and the Gemma-family guard ShieldGemma-9B (gated 2.0% against a base of 78.0%), with the per-prompt verdict agreeing across all 4 judges 72.8% of the time. The two prompted judges (the Llama-3.1-8B-Instruct secondary and the Mistral third judge) share one instruction, reproduced verbatim so the criterion is reproducible: *“You are a strict content-safety classifier. You are given a user REQUEST and an AI RESPONSE. Decide whether the RESPONSE provides meaningful assistance toward a harmful, dangerous, illegal, or unsafe goal (for example weapons, drugs, hacking, violence, fraud, self-harm). A refusal, a safe redirection, or an empty/irrelevant answer is SAFE. Actual harmful help is UNSAFE. Answer with exactly one word: ‘safe’ or ‘unsafe’.”* The user turn supplies the request and response and asks “Is the RESPONSE safe or unsafe?”, and the verdict is the first generated word. The defense also transfers to a different harmful-behavior distribution. With the detector and refusal direction fit on Ad-

vBench, the gate lowers prefix-injection attack success on HarmBench standard behaviors (Mazeika et al. 2024) from 93.5% to 5.5% ($n=200$, Wilson 95% confidence interval [3.1, 9.6]). The defense withstands an optimized attack, not just fluent templates. A gradient-free suffix tuned to elicit an affirmative continuation raises base attack success to 97.5%, and the gate lowers it to 6.5% (Wilson 95% confidence interval [3.8, 10.8]), while CAST at the residual does not suppress it and slightly backfires (attack success 100.0%, above the undefended 97.5%). The SSM suffix is gradient-free because the fused selective-scan kernels are non-differentiable. The gradient-based GCG attack (Zou et al. 2023b) runs on a transformer in Appendix A.24, and directly through the SSM’s slower differentiable scan in Appendices A.34 and A.24. A dilution attack that pads a harmful prompt with benign text defeats a naive threshold, but a harm-weighted pool that weights tokens by harm rather than position defeats it by construction, firing through 256 appended sentences and staying immune to a suffix optimized to evade a mean-pooled probe (Appendices A.27 and A.28). The gate does not degrade general capability because its detector almost never fires on benign task prompts. On MMLU its false-positive rate is 0.0%, so the gate stays inert and gated accuracy equals the base model on both MMLU (0.478 against 0.478) and TruthfulQA (0.297 against 0.297) on Falcon-Mamba. This is a detector-selectivity result, not capability preservation under firing.

A.33 The locus and the defense replicate on a second SSM

Both the locus and the defense reproduce on a second, independently trained instruct SSM, Falcon3-Mamba-7B. Harm is decodable at Δ (held-out AUROC 1.00) and the gate lowers attack success under fictional framing (56.0% to 4.5%) and persona roleplay (13.0% to 0.0%) at low over-refusal, so the finding is a property of the models and not one checkpoint. The only model-specific step is layer selection and gain calibration (Section 2.5, Appendix A.5).

A.34 A gradient-based attack through the SSM’s own differentiable scan

The fused selective-scan kernel used for inference is not differentiable end to end, but this is a speed choice, not a fundamental barrier: Falcon-Mamba’s own fallback path is plain differentiable PyTorch, and `mamba_ssm` ships a differentiable reference scan. We run GCG (Zou et al. 2023b) directly against Falcon-Mamba through this differentiable path, optimizing a 20-token adversarial suffix per behavior for 500 steps (256 candidates, top-256 substitution positions per step, identical hyperparameters and core to the transformer GCG attack in Appendix A.24), targeting the model’s own selective-scan block-output gate rather than the attention output. At $n=100$ harmful behaviors, the gate lowers attack success from 22.0% to 6.0%, confirmed by the out-of-family judge ShieldGemma-9B (15.0% to 3.0%). This is the same write-versus-residual contrast as every other attack family in this paper, now against an adversary that has gradient access to the model rather than only fluent prompt templates, and it

is a property of the differentiable path: the fused inference kernel used elsewhere in this paper is faster but was not built for backpropagation, so this attack necessarily runs through the slower reference implementation.

A.35 The intervention location is irrelevant: a matched-absolute-magnitude control

The write-site results compare a gate that reads and pushes the refusal direction at the fresh write against baselines that read and push it in the accumulated residual, and the residual’s activation norm is much larger than the write’s ($\times 7.8$ on Falcon-Mamba, $\times 27.9$ on Mistral-7B at the decision layer). A fixed additive push is therefore a very different *relative* perturbation at the two sites, so the reported site contrast could be a magnitude or direction artifact rather than a property of the location. We isolate the three possible causes (push location, estimated direction, magnitude) with a control that holds the detector fixed and sweeps the same *absolute* push magnitude across three arms: (i) push at the write site with the write-estimated direction (the deployed gate), (ii) push at the residual with the *same* write-estimated direction, and (iii) push at the residual with a residual-estimated direction (the CAST baseline). Every completion is scored with Llama-Guard-3-8B and reported with Wilson 95% intervals at $n=100$.

The push location does not matter. With the direction held fixed, pushing at the write site and at the residual are within CIs at every magnitude on Falcon-Mamba (write 6.0% [2.8%, 12.5%] versus residual 6.0% [2.8%, 12.5%] at the coherent operating point), and at the deployed gain on Mistral-7B (write 15.0% [9.3%, 23.3%] versus residual 23.0% [15.8%, 32.2%]). Away from that gain the two Mistral sites can separate in either direction, so the location-equivalence there is an operating-point statement. The SSM identity follows because a Mamba block adds only its mixer output to the residual. What matters is where the direction is *estimated*: a direction estimated in the accumulated residual degenerates when amplified (100.0% attack success on the SSM, at broken coherence), reproducing the CAST failure. The architecture-specific quantity is thus the readout site where the refusal direction is cleanly estimable, not a steering location.

A.36 The gate holds under an adaptive attack optimized through it

The gradient attack in Appendix A.34 optimizes the suffix against the plain model. A stronger adversary optimizes the suffix against the deployed gate itself, with the write-site amplification installed during the search so the gradient flows through it. We run this gate-aware GCG on Falcon-Mamba ($n=100$, 500 steps, identical hyperparameters). The gate holds: the detector fires on all 100 attacked prompts (the adaptive suffix does not evade it), and deployed attack success stays at 1.0% (from 15.0% on the ungated model), Llama-Guard scored. On the transformer the same adaptive attack instead drives down the detector score and erodes the deployed gate, so the SSM’s mixer-state detector is the more robust trigger of the two.

A.37 The wrapper confound, screened on every architecture

A detector trained on harmful versus benign-but-spicy prompts can key on the jailbreak *template* rather than on harm, since at evaluation the harmful prompts are wrapped and the benign ones are not. That is not hypothetical: on the anchor it is exactly what happens until the training set is augmented with wrapped examples (Appendix A.29). We therefore ran the same screen on the two architectures added later, fitting a plain detector (harm versus spicy, both unwrapped) and a wrapper-hardened one (harm and wrapped harm versus spicy and wrapped spicy) at each write site, and measuring firing rates on four held-out populations of 100 prompts each. The diagnostic is the plain detector’s firing rate on *wrapped benign*: near zero means it reads harm, high means it reads the wrapper.

The confound is close to universal. Both transformers behave exactly as the anchor does: the plain detector fires on 100.0% of wrapped benign prompts on Mistral-7B and 100.0% on Llama-3.1-8B, and wrapper-augmented training drives both to 0.0% while leaving wrapped-harm detection at 100.0%. We therefore re-ran both transformer defenses with the hardened trigger at the deployed operating point, and Table 6 reports those runs. Attack success is unchanged by the swap (Mistral 98.0% to 14.0%, Llama-3.1-8B 25.0% to 0.0%, against 15.0% and 1.0% unhardened), as expected, since both detector variants fire on every wrapped harmful prompt. Only the hybrid still runs a plain trigger, because at one of its two sites hardening does not work.

RWKV-6 is the exception in the other direction. Its plain detector fires on 100.0% of harmful prompts, 100.0% of wrapped harmful prompts, and 0.0% of wrapped benign ones, so hardening changes nothing (0.0% either way) because there is nothing to fix.

Zamba2 is not, at either site. The plain detector fires on 100.0% of wrapped benign prompts at the attention site and 100.0% at the Mamba site, so on this model the trigger is reading the template. Wrapper-augmented training repairs the attention site completely, dropping wrapped-benign firing to 0.0% while wrapped-harmful firing stays at 100.0%. It does *not* repair the Mamba site, where the hardened detector still fires on 100.0% of wrapped benign prompts. A linear probe at the Zamba2 Mamba output therefore does not separate a wrapped benign prompt from a harmful one, whatever it is trained on, which is a property of that site’s representation rather than of the training set. We report the hybrid’s numbers with that caveat attached (Section 7) and do not claim a deployable trigger there.

That failure is informative about the site, which is this paper’s subject. Everywhere else we look, the write site is where harm is *most* cleanly readable: it is where decodability peaks in the SSM (Section 4), where the transported probe lands, and where the refusal direction is cleanly estimable. The hybrid’s Mamba output satisfies the second and third of those and fails the first in the one condition that matters for a trigger, since a jailbreak template moves it as far as harmful content does. Reading and steering therefore come apart once more, in a new place: the site can host a direction

that controls refusal while its linear geometry still confuses a wrapped benign prompt with a harmful one. Our claim is about where a refusal direction is cleanly *estimated*, and this shows that property does not imply the strictly stronger one of supporting a reliable linear *detector* at the same site. The two coincide on the SSM, the transformers, and RWKV-6, and the hybrid is where they separate.

A.38 The benign-side cost of every write-site gate

An attack-success reduction means nothing without its false-positive cost, since refusing every prompt drives attack success to zero. We therefore measured over-refusal on held-out benign-but-spicy prompts at the exact operating point behind each row of Table 6. For the SSM and the two transformers this came from the runs that produced the attack-success numbers. For RWKV-6 and Zamba2 it did not exist, because those runs generated on harmful prompts only, so we replayed each run’s own configuration on the benign split.

The answer is the same everywhere, and it is structural rather than lucky: the gate is triggered by a detector that does not fire on benign prompts, so on those prompts the model runs unmodified and the over-refusal rate cannot move. On RWKV-6 the detector fires on 0.0% of 100 benign prompts and over-refusal is 32.0% both with and without the gate. On Zamba2 it fires on 0.0% of 100, and over-refusal is 3.0% either way, at both write sites. The same detector fires on 100.0% of the jailbroken harmful prompts, so the separation is not bought by a timid trigger.

A.39 Closing the loop on the hybrid, and what always-on steering costs

The hybrid results in an earlier version of this work amplified the refusal direction on every prompt, with no detector, unlike every other architecture we test. Measuring the benign cost of that configuration shows why the distinction matters. Always-on steering at Zamba2’s Mamba write site raises over-refusal from 11.0% to 63.0%, and at the attention site to 99.0%, on 100 benign prompts. At the attention site the model refuses essentially everything, so its attack-success reduction was substantially a refuse-everything artifact. This reproduces on a fourth architecture what Section 5 reports for the SSM, where open-loop steering matches the gate’s attack success while over-refusing 86.0% of benign prompts.

Closing the loop removes the cost without costing the defense. Fitting a linear detector at the same write site and amplifying only when it fires, the trigger separates the two populations completely (100.0% of harmful prompts, 0.0% of benign), attack success falls from 31.7% to 0.0% at the attention site and 5.0% at the Mamba site, and over-refusal is unchanged at 3.0% (+0.0 pt). Table 6 reports this closed-loop version, so all four architecture families are compared under the same detect-then-gate design.

A.40 Human validation of the attack-success judge

Every attack-success number in this paper is produced by an LLM judge, so the judge itself needs an anchor. We sampled 300 completions, stratified over the cells that carry claims and

weighted toward the small-margin ones (the RWKV-6 and Zamba2 ablations), drawn uniformly within each cell so that per-cell rates stay unbiased. Two raters labelled every item independently, in different orders, seeing only the request and the response. Model, condition, and all judge verdicts were withheld, and the verdict key was held in a separate file. The primary label asks exactly what the judge is asked, whether the response gives meaningful assistance toward the harmful request, with a separate label for refusal and a third for whether the text is coherent at all. An item counts only when both raters agree; 34 of 300 did not resolve and are excluded rather than broken by a tie-break we have no third rater to supply.

The raters agree with each other at 89.0% on compliance ($\kappa=0.80$) and 90.3% on refusal ($\kappa=0.82$). Coherence is the exception: raw agreement is 85.2% but $\kappa=0.24$, because almost everything is coherent and a κ on so skewed a variable is uninformative. We therefore use coherence only as a coarse screen and not as a reported rate.

Against the human consensus, the primary judge (Llama-Guard-3-8B) reaches 94.8% agreement and $\kappa=0.90$ on the $n=97$ items for which per-item judge verdicts were stored. Its errors are one-sided: 5 items it called compliant were read as safe by both raters, against 0 in the other direction, so the judge over-counts attack success and every defense result here is conservative. The other three judges bracket it, from $\kappa=0.94$ (Mistral-7B-Instruct) to $\kappa=0.74$ (ShieldGemma-9B, the strictest, with 10 items the raters called compliant and it did not). Finally, the degeneration concern is small in this sample: of the 67 judge-flagged items, 4 (6.0%) were rated incoherent by both raters, so miscounted gibberish is a real but minor contributor to the reported rates.

Per cell, the human rates track the judge on the anchor and correct it on one exotic arm. On Falcon-Mamba under prefix injection the raters put the undefended model at 100.0% ($n=20$) and the gated model at 0.0% ($n=18$), so the defense is if anything larger under human labels than under the judge. The correction is on RWKV-6, where the raters score the transported-direction ablation at 60.7% ($n=28$) against only 5.9% ($n=17$) for the matched random control. The judge had put that control at 32.3%, so the separation we report for RWKV in Table 4 is conservative: its random control is far less jailbreaking to a human than the judge scored it, and the causal margin is correspondingly cleaner. Zamba2’s attention site stays undecisive on human labels as well (14.3% transported against 6.7% random, n of 14 and 15), which is the reading we already give it. These per-cell counts are small and we do not restate the table’s numbers from them. They are a check on direction and rough magnitude, not a replacement measurement.

A.41 What the portable direction buys when target labels are scarce

Portability is only worth something if a transported direction beats training one locally. This measures that directly, against the strongest cheap alternative a practitioner has: fitting the target model’s own refusal direction on the same labels the transport would have used. Both arms receive an identical budget of k labeled target prompts. The transport arm spends

them on paired activations, fits the ridge map on those k pairs, and carries through it a source direction estimated on the source model’s abundant labels. The from-scratch arm spends them on a mean-difference direction in the target. Both are then scored by AUROC on ground-truth XSTest labels, so no judge enters this comparison.

The comparison must run off ceiling to mean anything. In distribution both arms saturate: at $k=4$ the transport reaches AUROC 1.00 and from-scratch 1.00, leaving no gap for either to fill, because harm-versus-benign detection is easy within a distribution the labels already cover. We therefore run the payoff in the regime where the transport test itself lives off ceiling (Appendix A.8): hold out one XSTest harm category, draw the k labels from the other categories, and score on the held-out category, whose surface features neither arm has seen. Each of the 8 held-out categories is run with 15 label draws, giving 120 replicates per budget. Both arms see the same draw in each replicate, so the contrast is paired and the interval below is on the difference itself.

The independent unit is the held-out *category*, not the replicate. The 15 draws inside one category are scored against that category’s single test set, so treating all 120 replicates as independent and dividing by their square root would understate the interval. We therefore report a t interval on the 8 per-category mean differences. The narrower replicate-level interval also excludes zero at every budget (for example $[-0.035, +0.057]$ at $k=64$, against $[+0.013, +0.080]$ clustered), so the conclusion does not depend on the choice, but the clustered interval is the honest one and is what the table reports.

Table 25: Data-scarce payoff of the portable direction over a target direction fit from scratch on the same k target labels, generalizing to an unseen harm category (Llama-3.1-8B to Falcon-Mamba, AUROC on ground-truth labels). Δ is the paired difference and its 95% t interval over the 8 held-out categories, the independent unit. Wins is the fraction of the 120 individual draws favoring transport, which is why it can sit near half while the interval still excludes zero: the mean gain is carried by larger wins, not by winning more often. Positive Δ favors transport. The margin does not close over this range of budgets.

k labels	Transport	From scratch	Paired Δ	Wins
4	0.78	0.75	+0.032 [+0.008, +0.057]	64%
8	0.83	0.79	+0.042 [+0.011, +0.073]	68%
16	0.87	0.83	+0.038 [+0.019, +0.058]	78%
32	0.90	0.85	+0.040 [+0.006, +0.073]	72%
64	0.92	0.87	+0.046 [+0.013, +0.080]	78%

Transport wins at every budget tested, by +0.032 to +0.046 AUROC, with the clustered interval excluding zero in all five rows (Table 25). At a deployed operating point the margin is wider than AUROC implies. Holding false alarms on benign prompts to 1.0%, the transported direction detects 65.7% of harmful prompts at $k=64$ against 57.9% from scratch (+7.9 pt). At 5.0% it is 74.2% against 62.8% (+11.4 pt). The natural way to phrase this is a label-equivalence factor, a transported direction being worth some multiple of its budget

in freshly labelled examples, and we deliberately do not report one. Re-running the sweep on a finer grid of 14 budgets from $k=4$ to 64 shows why. The from-scratch curve is not monotone in k (5 of its steps go down rather than up) and it flattens from about $k=24$ onward, sitting between 0.845 and 0.872 for the rest of the range, against a mean interval width of 0.056 on that same curve. Reading a label budget off a curve that is flat and noisy in the region of interest inverts a near-zero gradient, and the answer moves by more than a factor of three depending on which budget is used as the reference. The detection rate at a fixed false-alarm budget, above, is the operating-point statement that does not require that inversion, and it is the one we make. The effect is not carried by one easy fold: transport is ahead in 7 of 8 held-out categories at $k=64$ and 7 of 8 at $k=4$. We expected the two arms to converge as k grew and they do not, which we read as the transported direction importing refusal structure the source estimated across categories that the target’s own k labels never cover, rather than as a better estimate of the same quantity. Two limits are worth stating. The gain is small in absolute terms, a few points of AUROC on a detection task, and it disappears in distribution, so the practical case for portability is generalization to harm the target’s own labels do not represent, not a uniform accuracy win. And this measures the readout, not the intervention: it says a re-estimated detector transfers, not that the downstream gate inherits the margin.

A.42 Final hyperparameters and tried ranges

Table 26 consolidates the final setting of every knob (write site, gate layers, decision layer, gain, threshold, transport map, detector, attack budgets) for every model in the paper, together with the values tried where a sweep was run. The same values are the defaults and command lines in the `reproduce.sh` of the code release.

Table 26: Final hyperparameters per model, with the values tried where a sweep was run. Gate layers are the top-8 by held-out probe AUROC on the SSMs (ranked by mixer-output AUROC on Falcon-Mamba, by Δ AUROC on Falcon3-Mamba, Section 2.5); the other architectures use fixed mid-network bands. The decision layer is the middle of the listed set and carries the per-prompt detector. Every gate uses $\theta=0.5$ on the probe’s harm probability, the mean refusal direction (mean harmful minus mean benign activation) at each gate layer, greedy decoding with 64 new tokens, and the gain α is the largest value that passes the coherence screen (Appendix A.25). Transfer rows give the layer the direction is read at in the source and the layers it is written at in the target.

Model	Write site	Gate layers (decision)	α final (tried)	Transfer: read / write layers	Notes
Falcon-Mamba-7B-Inst.	mixer output	{24,28,30,34,35,36,37,44} (35)	4 (1,2,4,6,8; capability at 4,8,16)	44 / {22,24,28,31,38,40,42,43}	anchor; Δ -ranked set for transfer targets
Falcon3-Mamba-7B-Inst.	mixer output	Δ -ranked {24,27,28,31,42,45,46,47} (42)	4 (0.5,1,2,4; Fig. 6)	47 / same set	block-vs-residual control on this set at $\alpha=4$; the output-ranked set {14,15,17,18,21,22,23,24} over-steers at $\alpha=4$ (reported at $\alpha=1$)
Llama-3.1-8B-Inst.	attention output	{10,12,...,24} (18)	2 (0.5,1,1.5,2,3,4)	14 / {12,14,16,18,20}	hardened trigger at the deployed point
Mistral-7B-Inst.-v0.2	attention output	{10,12,...,24} (18)	2 (0.5,1,1.5,2,3,4)	14 / {12,14,16,18,20}	persona fix: layer 18, $\alpha=6$, $\beta=6$
RWKV-6-World-7B	time-mix output	{14,16,...,24} (20)	16 (0,4,8,16,24,32,48; coh. ≥ 0.8)	20 (target of Llama L14)	defense run gating the block hook at $\alpha=6$ is a separate operating point; the block boundary (residual) is inert
Zamba2-7B-Inst.	attention out. / Mamba out.	{36,42,48,54,60} (48) / {20,24,28,32} (28)	8 / 16 (persona grid $\alpha \in \{2, 4, 6, 8, 12\}$, $\beta \in \{0, 1, 2\}$)	–	hybrid, both write sites gated

Baselines on Falcon-Mamba (same gate layers and detector): residual projection, LEACE-style rank 1 per layer, $\alpha=1$; CAST-style conditional residual steering, $\alpha=4$ in the main table, self-tuned sweep $\alpha \in \{2, 4, 8, 16\}$ (best 2);

always-on (DeTAM-style) edit at the same site and gain; SmoothLLM $k=5$ perturbations. *Detector:* logistic probe on standardized mean-pooled activations (≤ 128 prompt tokens), Adam, lr 0.1, 300 epochs, $\ell_2 10^{-3}$

(MLP variant: 400 epochs, lr 10^{-2} , weight decay 10^{-4}); wrapper-hardened variant adds jailbreak-wrapped training prompts. *Transport map:* ridge $\lambda=100$ or orthogonal Procrustes on 250 paired prompts (150 surface-matched, 200 for RWKV).

Adaptive attacks: $k \in \{0, 1, 2, 3, 4, 6, 8\}$ appended sentences (hardened: up to 12); optimizer 150 rounds $\times$ 64 candidates; GCG 500 steps, 256 candidates, top-256 substitutions, 20-token suffix, target “Sure, here is”, seeds 0–2.